

# **Macro-eco-evolutionary perspectives on the Cenozoic history of South American Mammals**

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# **Préambule et Remerciements**

**Avant-propos**

**Remerciements / Acknowledgements / Agradecimientos**

## Résumé étendu en Français

Cette section a pour vocation de produire un bilan un tant soit peu intelligible et digeste des circonvolutions intellectuelles autour de l'histoire évolutive des mammifères sud-Américains ayant animé mes trois dernières années, dans la langue de Molière.

### Considérations préliminaires

En premier lieu, une justification s'impose: pourquoi, en tant que scientifique européen, s'intéresser à la biodiversité d'un continent si lointain ? Je vous invite pour cela à vous référer à la carte présentée dans la Figure 1.2 de l'introduction. Cette carte représente, à l'actuel, le nombre d'espèces par unité de surface (aussi appelée richesse spécifique) chez les mammifères terrestres (A), les oiseaux non-marins (B) et les amphibiens (C). Un contraste frappant s'impose alors à nous: l'Amérique du Sud se démarque par son immense richesse spécifique, quasiment supérieure à n'importe quelle autre région du globe. En cela, il est coutume de qualifier le continent sud-Américain (et tout particulièrement sa partie tropicale, en première ligne de laquelle se trouve la fameuse forêt Amazonienne, voir Figure 1.8) de *hotspot* (pouvant se traduire par “haut-lieu”) de biodiversité (Raven *et al.* 2020).

L'origine de ce déséquilibre notoire de biodiversité à la surface du globe n'a rien de trivial, et a intrigué des générations de scientifiques depuis les travaux fondateurs d'Alexander von Humboldt (*p.ex.*, Humboldt & Bonpland 1807), Alfred Russel Wallace (*p.ex.*, Wallace 1876, 1878) ou encore du célèbre Charles Darwin (*p.ex.*, Darwin 1859).

Se surimpose à cela le contraste extrêmement marqué entre la biodiversité sud-Américaine actuelle et celle retrouvée dans le registre fossile.

Pour tout cela, l'histoire de la biodiversité mammalienne sud-Américaine regorge de mystères, dont beaucoup demeurent encore irrésolus à l'heure actuelle. C'est dans ce sillage que se range ce travail doctoral.

# 1 General Introduction

## 1.1 The unevenness of biodiversity: corner stone for the large-scale study of life

The astonishing **diversity** of life on Earth has fascinated scientists for centuries. A study dating back to 15 years ago indeed estimated that around 8.7 million eukaryotic species are currently coexisting on our planet (Mora *et al.* 2011). This number would probably be significantly increased by accounting for the substantial improvements that have occurred in the field of systematics over the recent years, and expanded by orders of magnitude if prokaryotic species were incorporated. Yet, someone travelling to the Sahara desert would probably be less amazed by the surrounding diversity of life than someone travelling to the middle of **the Amazonia**, one of the most species-rich regions in the world (Raven *et al.* 2020). Another vertiginous **data** is that around 99.9% of the species that ever existed since the dawn of life on Earth, some four billion years ago, has gone extinct (Raup 1986). It therefore stands out that, far from a fixed and smooth entity, biodiversity, here understood as species richness, presents a myriad of heterogeneities, that stimulated a vast quantity of research works in which this thesis happens to be nested.

### 1.1.1 The heterogeneous organisation of life across space

For more than two centuries, it has been recognised that species are not randomly distributed on Earth. In his pioneering “*Essai sur la géographie des plantes*”, regarded as foundational for the science of Biogeography (see **Glossary**), Alexander von Humboldt emphasised the spatial nature of plant taxonomic assemblages (Humboldt & Bonpland 1807). Later in the 19<sup>th</sup> century, the scientific community acquired a uniquely integrated perspective on biodiversity fed by years of field observations, giving to the question of the spatial organisation of life on Earth a new dimension. Consequently, it has been proposed to subdivide the world into spatial units sharing similar fauna/flora, sometimes called ‘Homozoic Belts’, ‘Regions’, ‘Realms’, ‘Provinces’, ‘Areas’ or ‘*Gebiet*’ (German word for territory, area) (Burkill 1943). The history will mainly remember the efforts of Alfred R. Wallace in that respect, who relied on the subdivision made by Philip L. Sclater in 1874 based on mammals, as well as his own expertise, to establish the famous contouring of the Earth’s ‘Zoogeographic Realms’ (Figure 1.1), two years after (Wallace 1876). Further efforts towards incorporating fossil evidence to better constrain the aforementioned subdivisions have to be acknowledged, notably implused by Richard Lydekker (Lydekker 1896). Wallace’s subdivision has maintained in the field since its publication, and only received minor updates in the last decade (Holt *et al.* 2013), illustrating the degree of knowledge advancement of the English naturalist, in a context where **Charles Darwin’s famous Theory of Evolution** – that, actually, Wallace co-discovered in a totally independent manner, making such a theory one of the most striking examples of thought convergence in the history of sciences (Berry & Browne 2008) – still had a long way to go before being formally accepted by the broad scientific community.

Beyond its spatial organisation into units of shared taxonomic composition, biodiversity also ex-



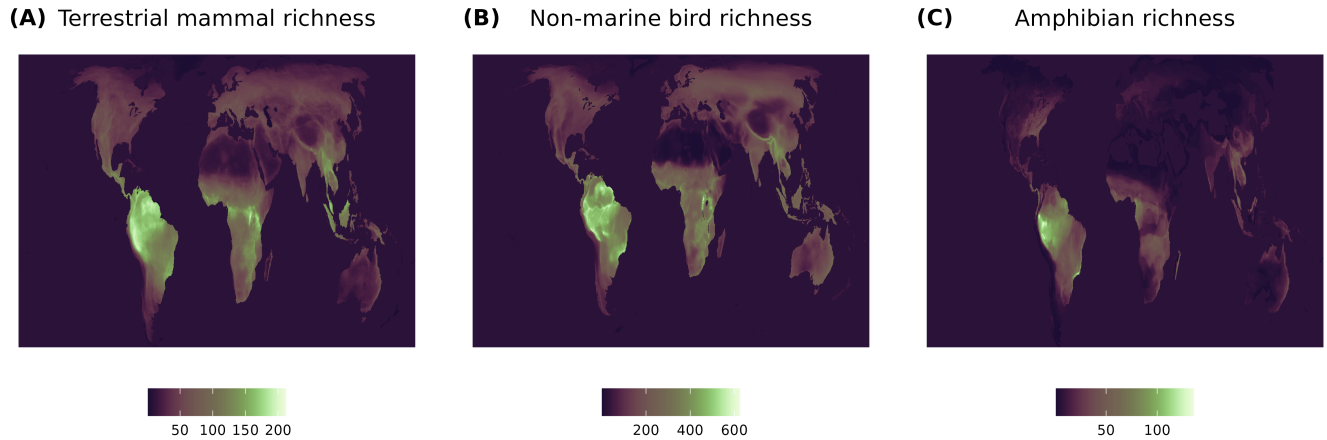
Figure 1.1: Biogeographic map based on faunal similarities within animal assemblages across space ('Zoogeographic Realms') proposed by Wallace (1876).

hibits a striking latitudinal organisation, in which species richness increases with decreasing latitude, known as the Latitudinal Biodiversity Gradient (Figure 1.2). Such a pattern was already recognised by Humboldt. According to him, low-latitude plant tropical assemblages would be essentially made up with a large number of different species per unit of area, each having a relatively low density ("isolated and sparse"). Conversely, higher-latitude temperate or cold assemblages would be characterised by a substantially lower species richness per unit of area, with a single dominant species being over-represented ("excluding [...] heterogeneous species") (Humboldt & Bonpland 1807). Half a century later, in his cornerstone book "*The Origin of Species*", Darwin dedicated two chapters to the spatial distribution of species, recognising that many groups harbour a greater diversity in the tropics and seeking for potential mechanisms underlying this pattern (Darwin 1859). Today, though notable exceptions exist (*e.g.*, pinnipeds, lagomorphs or several kinds of fungi), a latitudinal diversity gradient has been recognised among most taxonomic group (*e.g.*, plants, insects, molluscs, vertebrates) (Hillebrand 2004), as depicted on Figure 1.2 for three vertebrate classes.

Hence, biodiversity is largely heterogeneous across space, raising questions that have been thoroughly investigated in the past, as to 'Why are there more species in the tropics than in the poles?' or, more generally, 'What constraints the distribution of species on our planet?'.

### 1.1.2 From peaks to declines: temporally-heterogeneous biodiversity

The aforementioned complex spatial arrangement of life is in fact the legacy of hundreds of millions of years over which **species have been fashioned** by a wealth of forces – of which we intend to provide an overview in Section 1.2.1. The statement that species evolve across time, and that species-level temporal changes eventually scale up to multispecies assemblages making up biodiversity is now commonly accepted, mostly thanks to fossil data.



**Figure 1.2:** Extant land mammal (A), non-marine bird (B) and amphibian (C) species richness at the surface of the globe, expressed in number of species, with a 10 x 10km resolution. Data are from <https://biodiversitymapping.org/>

Being the only source of direct evidence of life in the past, fossils have indeed been of immense value to characterise the temporal dynamics of biodiversity, notably since they irrefutably document the occurrence of extinction in the history of life. It is however interesting to notice that the reasoning that makes us conclude that if, for instance, mammoths are found in the fossil record but not among extant assemblages, it is because they have gone extinct, has not always been as straightforward as it appears to us. Fossils have indeed puzzled eminent figures in the field of biology, such as **Jean-Baptiste de Lamarck – to whom we incidentally owe the name ‘biology’**. In the early 1800s, Lamarck tentatively formalised a theory where animals would be subject to a complexifying force driving body plans towards higher levels. According to his theory, the (animal) biosphere would thus represent a continuum where species would be unidirectionally connected between each others, which, importantly, leaves no room for extinction – or any sort of species richness variation across time – : every member of Lamarck’s gradual sequence towards increasingly-complex body plans would already exist ([Lamarck 1815](#)). However, Lamarck had to cope with the fact that the field of systematic palaeontology was emerging in parallel, notably impelled by his colleagues from the Muséum National d’Histoire Naturelle de Paris, including Georges Cuvier ([Servais \*et al.\* 2012](#)). Plenty of remains of animals trapped inside the rocks but also manifestly absent from the surface of the (known) world were thus unearthed and described, challenging Lamarck’s model. In order to make sense of this apparent inadequacy, Lamarck formulated a famous *ad-hoc* hypothesis: the apparent absence of extant representatives of mammoths does not mean that mammoths have gone extinct. If they indeed locally disappeared from the places where they left fossil remains, they must have found refugia in a region of the globe unexplored by the occidental civilisation.

Markedly opposed to Lamarck’s theory of animal organisation following increasingly complex body plans, Cuvier was maybe one of the earliest scientists to propose that species assemblages could change across time, although the theory that he developed (‘catastrophism’) was largely invalidated later in the history. His rich fieldwork expertise indeed led him to postulate that the history of the globe has been punctuated by abrupt ‘revolution’ events, which can be viewed as (abiotic) changes occurring within a given environment, in turn causing changes in the nature of the species present within this changing environment ([Cuvier 1840](#)). Two decades later, John Phillips, an English palaeontologist notably known for giving Phanerozoic eras their name (Palaeozoic, Mesozoic and Cenozoic) compiled

for the first time fossil occurrences of marine taxa from the British lithological record, that he used to produce the first continuous diversity-through-time curve, spanning the entire Phanerozoic (Phillips 1860). His work allowed him to discriminate what he called the Palaeozoic, Mesozoic and Caenozoic ‘lives’. These three ‘lives’ were separated by vast depressions, corresponding to the Permo–Triassic and Cretaceous–Palaeogene boundaries (Figure 1.3 A), that will be later identified as two amongst the five most significant extinction episodes that marked the history of life, also known as mass extinction crises (see Section 1.2.1). Phillips’ cornerstone compilation thus, for the first time, quantitatively demonstrated the heterogeneity of species richness across geological times.

A century later, the rise of quantitative palaeobiology (see Section 1.1.4) paved the way towards more accurate reconstructions of biodiversity across time. The publication, in the early 1950s, of the first edition of the *Treatise of Invertebrate Paleontology* provided an access to a vast quantity of marine invertebrate fossil data, that could be further used for such an achievement. In that framework, Norman Newell reconstructed diversity curves for about 16 different groups (Newell 1952). Again, the reconstructed curves displayed time-varying patterns. Newell noticed certain similarities in the reconstructed patterns between groups, that he suggested to be evidence for “common causes” underlying their large-scale evolutionary trends. A few decades later, with a higher-resolution curve, Valentine (1969) (Figure 1.3 B) proposed to “explain diversity trends in terms of ecological hierarchy” (Sepkoski 2012), constituting a significant advance towards integrating ecological and evolutionary processes (see Section 1.2).

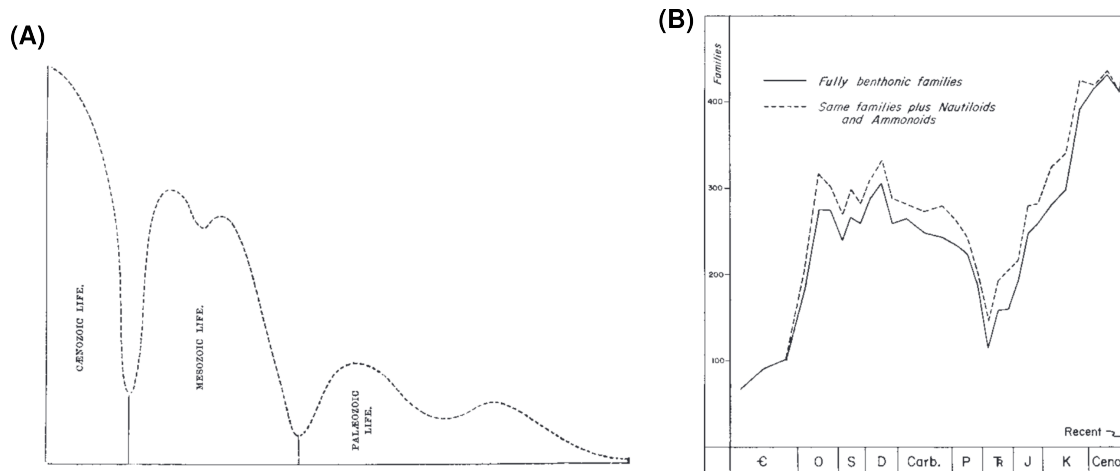


Figure 1.3: Historical fossil-based reconstructions of marine diversity across geological times produced by Phillips (1860), using a marine invertebrate compilation from the British lithological record (A) and by Valentine (1969), using marine invertebrate record from the *Treatise of Invertebrate Paleontology* (B). Note that timescale in (A) has been reverted when compared (B), going back in time from left to right instead of right to left.

However, Valentine’s reconstructions raised some methodological criticism, especially expressed by David M. Raup, one of the leading figures of the ‘palaeobiological revolution’. Raup was indeed concerned about how the fossil record biases may affect Valentine’s reconstruction, importantly stressing that “the increase in the number of marine species across the Phanerozoic may be more apparent than real” (Raup 1972, see Section 1.1.4). Ten years of intense data compilation and methodological labour after, together with J. John Sepkoski Jr. (another major actor of the ‘palaeobiological revolution’) Raup came to the conclusion that marine biodiversity indeed experienced temporal variations, persisting even after accounting for biases inherent to the fossil record (Sepkoski *et al.* 1981). A year after,



Raup & Sepkoski (1982) characterised major temporal features in their diversity curve (Figure 1.4), now known as the ‘Big Five mass extinctions’ (that we will develop in further details in Section 1.2.1).

Nowadays, it makes no doubt that biodiversity is temporally heterogeneous. Clades originate, thrive and then possibly decline or even go extinct, sometimes fluctuating in species number in between. A core aim of macroevolution, in which this doctoral work is nested, is thus to elucidate, given a clade, how species diversity changes across time, and the factors underlying these changes (Section 1.2.1).

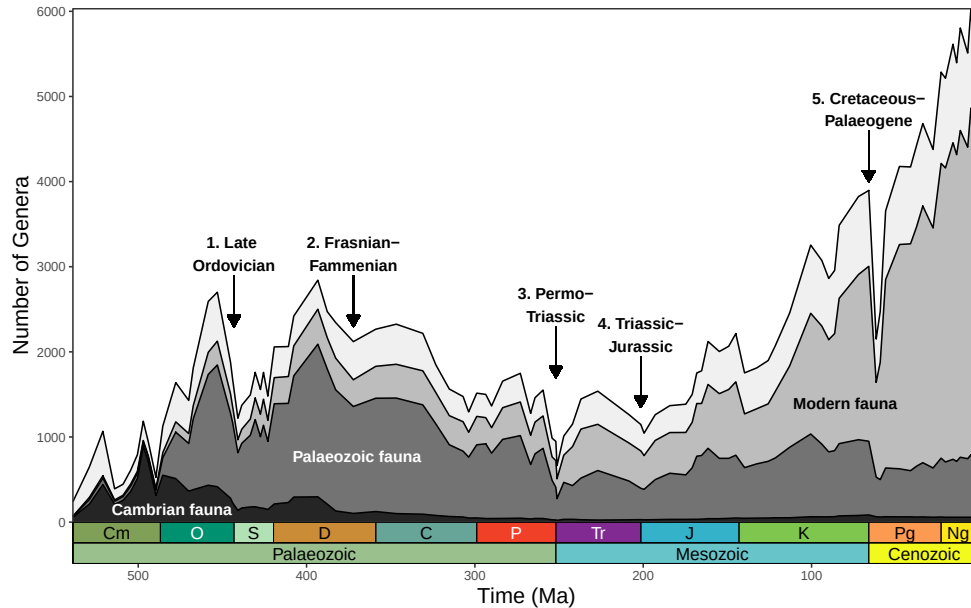


Figure 1.4: Temporal variations in marine animal biodiversity across the Phanerozoic. Colours indicate the three evolutionary faunas discriminated by Sepkoski (1984). Taxa that could not be assigned to any fauna are included in the clearest band. Arrows indicate the ‘Big Five’ mass extinctions, as defined by Raup & Sepkoski (1982). Data are from the genus-level compendium released by Sepkoski (2002), binned across geological stages, accessed and displayed thanks to the R package `sepkoski` (Jones 2022). Geological timescale was added thanks to the `deeptime` R package (Gearty 2023).

### 1.1.3 Inter-clade heterogeneity

Another facet of the heterogeneity of biodiversity comes from the fact that some groups harbour an immensely larger species richness than others. This striking feature has intrigued evolutionary biologists since the times of Darwin. In a 1879 letter to Joseph Hooker, the renowned white-bearded English naturalist indeed famously termed the astonishing diversity of flowering plants (Angiosperma) and their apparent rapid burst in the fossil record as an ‘abominable mystery’. At present, with more than 300,000 described species, angiosperms encompass nearly 90% of the extant land plant diversity.

Over the past few decades, the rise of phylogenetics (Section 1.3.1) permitted to better constrain evolutionary relationships between lineages, thus highlighting inter-clade diversity disequilibria (Figure 1.5). Understanding what drives striking diversity disequilibria between groups, even sometimes closely related, is a central aim of macroevolutionary research.

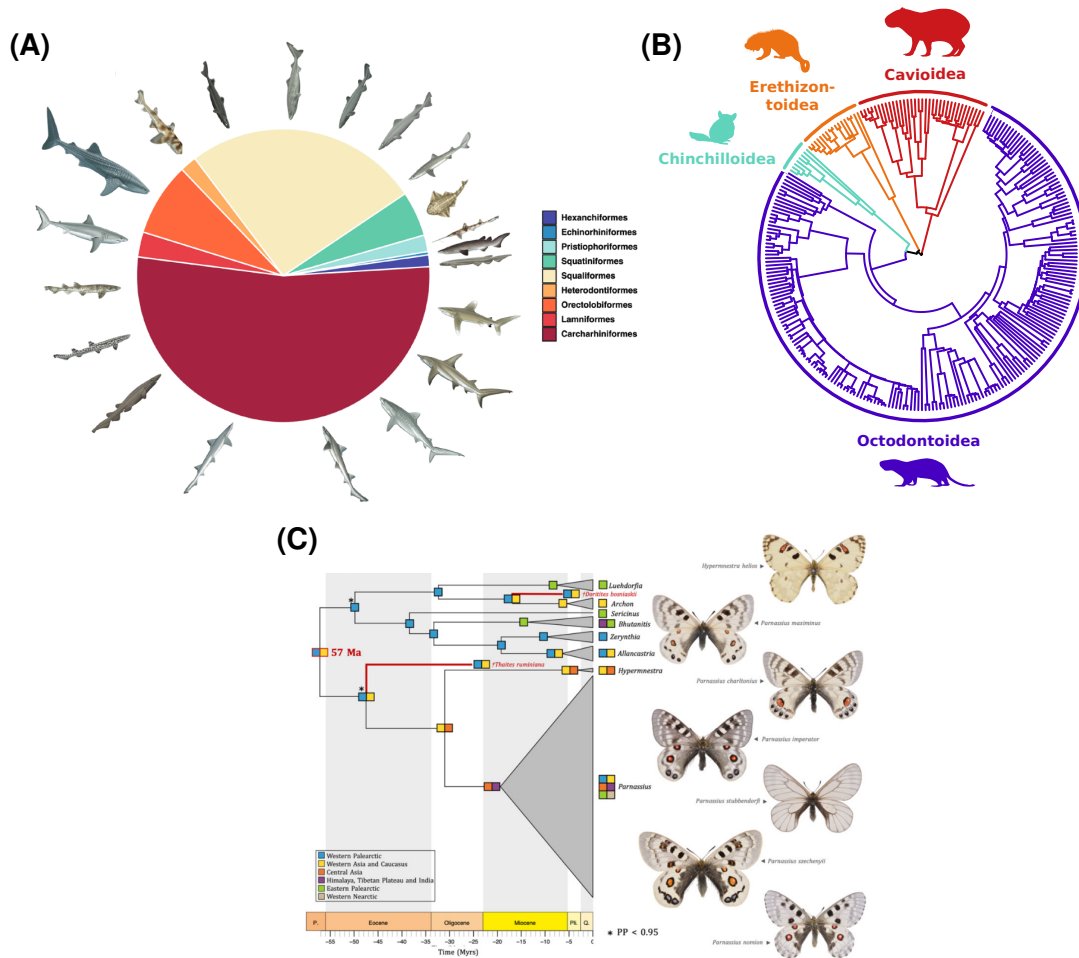


Figure 1.5: Examples of diversity disequilibria in the animal tree of life. **(A)** >50% of the extant shark (Neoselachii) diversity is comprised within the order Carcharhiniformes, whereas other orders (*e.g.*, Echinorhiniformes, Hexanchiformes) are comparatively way less diversified (figure reproduced with permission from Marion *et al.* (2024)). **(B)** ~80% of the extant caviomorph (Rodentia: Hystricognathi) diversity is comprised within the superfamily Octodontoidea, whereas the Chinchilloidea (see extcolor{Blue}{Chapter 2}) only comprises 8 living species. Phylogeny is from VerLife, and silhouettes are from Phylopic (Credits : Michael Tripoli and Gabriela Palomo Muñoz). **(C)** ~90% of the extant diversity of Parnassiinae (Lepidoptera: Papilionidae) is comprised within the genus *Parnassius* (figure reproduced with permission from Hévin & Condamine (2024)).

### 1.1.4 Methodological biases arising when studying biodiversity

#### Biodiversity shortfalls

Several caveats apply to the study of biodiversity patterns ‘from a distance’, the baseline of the macroecological perspective (Diniz-Filho 2023). Some of them are grouped into what is known as the ‘biodiversity shortfalls’, representing inherent limitations related to missing information in biodiversity



data, thereby inevitably hampering accurate understanding of the modelled eco-evolutionary process. Three biodiversity shortfalls are usually considered, though Hortal *et al.* (2015) in fact more finely discriminated seven – that will not be extensively covered here. These three have been named after three renowned figures of ecology, evolution and systematics.

The *Linnean shortfall* (named after Carl von Linné, the Swedish father of **systematics**) reflects the  $\alpha$ -taxonomic uncertainty, that is, the degree of certainty that one may have, after sampling two individuals at random in a community, to tell whether they belong to the same species or to two distinct species. Materialisations of the Linnean shortfall are, for instance, cryptic diversity (under-estimation of the number of species within a clade due to a lack of data and/or morphological similarities), or ‘taxonomic splitting’ (over-estimation of the number of species within a clade, often arising from the use of morphological criteria; it therefore particularly applies to the fossil record). Next, the *Wallacean shortfall* (named after Alfred R. Wallace, to whom this thesis will refer several times) refers to the knowledge gaps in species distribution, *i.e.*, the fact that a species may be somewhere where it has not been observed; thus this place in particular would be excluded from upstream range map reconstructions of that species (sort of type-1 error applied to biodiversity data). Last, the *Darwinian shortfall* (named after Charles Darwin, also mentioned several times across this thesis) **refers to the lack of knowledge of the evolutionary relationships between species**.

Additionally, the study of biodiversity has long been affected by cultural and/or geopolitical constraints, either limiting the broad accessibility of published knowledge (*e.g.*, written in non-latin alphabets) or the access to specimens and/or sampling localities.

### Propagation of the caveats of the fossil record when studying biodiversity through time

*“A single answer to the question of whether the fossil record is driven by macroevolution or megabias is unlikely ever to emerge because of temporal, geographical, and taxonomic variance in the data.”* (Benton *et al.* 2011)

Additional biases emerge from reconstructions of biodiversity patterns over long time scales, overlaying on the previously described biodiversity shortfalls. These reflect the inherent incompleteness of the fossil record.

Darwin himself emphasised the imperfection of the fossil record (somewhat understandable given how poor the fossil record was at his time compared to the present-day), comparing the latter to a book written in different dialects with many missing pages, and with only a few lines preserved on the remaining pages (Darwin 1859). This conception of the fossil record, too sparse to be accurately used to reconstruct evolutionary processes (“Darwin’s dilemma”, Sepkoski (2012)) largely spread among evolutionary biologists. Consequently, **during most of the 20th century, palaeontology was marginalised from the rest of the evolutionary biology community**. Roughly between 1970 and 1985, a massive change of paradigm occurred as some palaeontologists started to focus on large-scale analyses of evolutionary patterns through quantitative analyses of the fossil record, known as the “palaeobiological revolution” (Sepkoski 2012). Increasing attention was dedicated to fossil-related biases, giving birth to historical controversies about the way to “read” through the fossil record, as the famous one that opposed James W. Valentine and David M. Raup (Raup 1972; Valentine 1969). This period thus paved the way towards significant methodological development to infer processes related to the temporal dynamics of biodiversity accounting for several known limitations of the fossil record, yielding unprecedented findings calling many established paradigms on the dynamics of biodiversity into question (*e.g.*, Alroy *et al.* 2001; Barrett *et al.* 2009; Benson *et al.* 2016; Close *et al.* 2020; Smith

& McGowan 2007).

Essentially, fossil-related biases can be grouped into three main classes. The first one naturally relates to the exceptional phenomenon of *fossilisation*, operating a selective transition from stacks of organic matter to permanent imprints trapped inside the rocks (Cook *et al.* 1961). Fossilisation requires a very special array of environmental conditions that will synergetically interact to by-pass the almost inevitable process of organic matter decomposition (*e.g.*, rapid burial, anoxic conditions). Its selective nature has several facets. The *structural selectivity* of fossilisation lies within the fact that, provided the aforementioned environmental conditions are met, mineralised tissues (*e.g.*, teeth, bones) will be more easily preserved than soft tissues, the latter being almost never preserved (except in very exceptional cases, Shields (1998)). Furthermore, the *ecological selectivity* of fossilisation has to do with organisms' ecological preferences towards zones of high fossilisation potential. Consequently, fossilisation is an extremely rare and selective process.

The second class of biases arising from the fossil record is more human-related, as it involves the *sampling* process. Indeed, unearthing a fossil has some parts of contingency. Should a hypothetical excavation be conducted a few metres away from a spot hosting a highly-valuable fossil, such a fossil would probably never be found. That being said, far from being uniformly distributed at the surface of the globe, sampling effort exhibits a striking spatial structure (Smith 2001), being extremely high in some regions (*e.g.*, North America, Europe, Japan, Argentina), and extremely low in others (*e.g.*, tropical non-Andean northern South America, Africa, South East Asia, Siberia, Antarctica, Western Australia) (Figure 1.6). It is interesting to notice that the most species-rich regions at present (*i.e.*, the tropics, Figure 1.2) are often the most fossil-depauperated (Figure 1.6). Consequently, the *known fossil record* (*i.e.*, all the specimens that have been unearthed to date) is only a subsample of the *potential fossil record*, (*i.e.*, all the biological remains that came to fossilise in the history of life), itself a subsample of all the life that ever existed (Benton *et al.* 2011). The spatial structure of the fossil record is an ongoing research avenue, that led to several keystone discoveries (*e.g.*, Antell *et al.* 2024; Close *et al.* 2017; Flannery-Sutherland *et al.* 2022; Reijenga & Close 2025).

For a given taxon, the stochastic character of both sampling and fossilisation may lead to a significant truncation of its temporal persistence. In that case, the oldest and youngest occurrences of this taxon may fail to accurately reflect its actual origination and extinction ages, incorporating a significant bias to any upstream occurrence-based analysis. The importance of this bias raised particular concern for the detection of mass extinction events (Section 1.2.1), since many of the lineages actually going extinct during such events would be expected to disappear from the fossil record prior to its realisation. These artificially truncated age ranges would therefore tend to highlight a gradual extinction pattern starting well before the actual extinction, which goes in contradiction with the abrupt character of mass extinctions (see Glossary). This well-known bias has been termed ‘Signor-Lipps’ effect, from the names of the two authors first describing it (Signor & Lipps 1982).

The third class of biases can be referred to as time-related biases. These comprise the difficulty to replace a fossil in an absolute temporal framework (Behrensmeyer 2026). Indeed, dating a fossil is far from being a trivial task. In the best scenario, nearby absolute anchoring points (*i.e.*, minerals usually from tuffs or volcanic zircons that can be robustly dated with isotopes) allow to constrain the interval when the fossil-bearing beds were deposited (*absolute dating*). However, many deposits lack such temporal anchors, and are thus dated using indirect methods, that will not be detailed here (*relative dating*). Hence, age intervals coming along with a fossil can be large, which may in turn add another layer of uncertainty to upstream biodiversity-through-time reconstructions. Another source of time-related biases is the fact that the probability for a fossil to get eroded, and thus, lost, increases with its age. Consequently, the number of available fossil occurrences is expected to be larger for

younger assemblages, an effect termed ‘Pull of the Recent’ ([Foote 2000](#)).

Thus, studying large-scale biodiversity patterns across time requires embracing many sources of uncertainty.

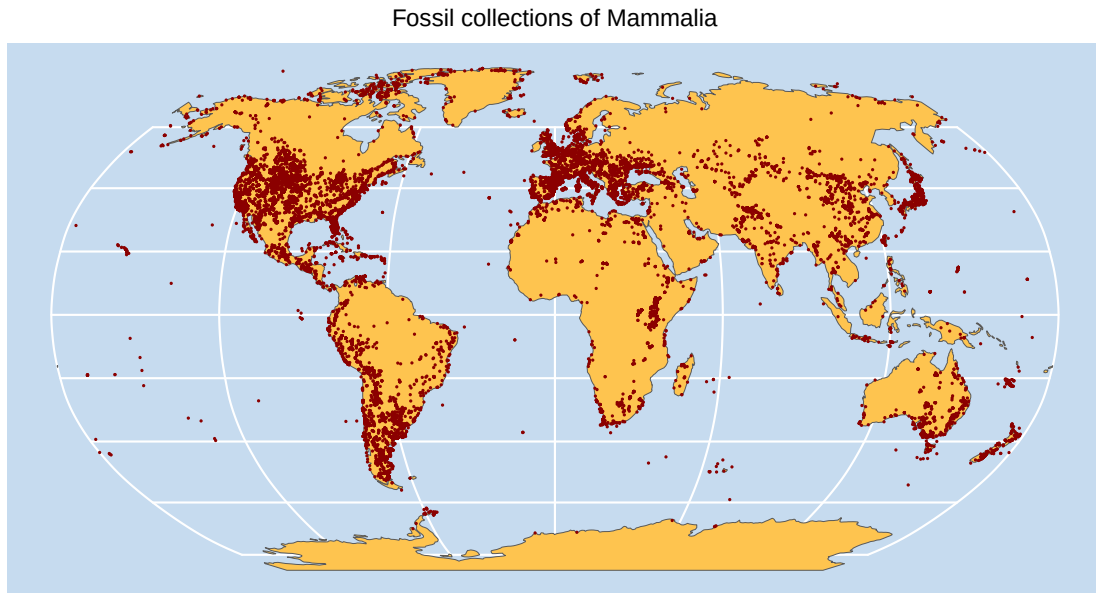


Figure 1.6: The spatial **heterogeneity** of sampling effort, here exemplified by the global distribution of mammal fossil collections (all geological periods confounded). Data from [The Paleobiology Database](#).

## 1.2 Disentangling biodiversity patterns across space and time: the macro-eco-evolutionary perspective

### 1.2.1 Tempo and mode of macroevolution

Macroevolution is usually understood as the study of evolutionary patterns occurring above the species level ([Hautmann 2020](#); [Saupe & Myers 2021](#)). It has been classically approached under two non-mutually exclusive lenses ([Stanley 1979](#)), either focusing on (1) the diversification process ([Glossary](#)), or (2) the evolutionary dynamics of phenotypes – as an echo to the term ‘diversification’, some authors refer to the process underlying morphological diversity as ‘disparification’ (*e.g.*, [Davis \*et al.\* 2016](#); [Nürk \*et al.\* 2019](#)). Unfortunately, despite their undeniable appeal, questions related to the dynamics of morphological evolution fall a bit outside the scope of this thesis. Thus, in this manuscript, emphasis will be placed on clade diversification (see [Glossary](#)), that is, the balance between the speciation and the extinction processes. The heterogeneity of biodiversity can be indeed viewed as the result from variations in the pace at which lineages originate and disappear, namely, extinction and speciation rates. Deciphering when (*tempo*), how and why (*mode*) these rates vary across space, time and clades is a core aim of macroevolution ([Saupe & Myers 2021](#)).

## Tempo of Macroevolution: when do rates change across time?

As seen in the previous section, the history of life has been punctuated by important fluctuations in diversity, possibly heterogeneous across lineages and geographical areas. Marked increases in diversity, either generated by an increase in speciation rate or a decrease in extinction rate (or a combination of the two), known as evolutionary radiations, have long been recognised (*e.g.*, Losos 2009; Simões *et al.* 2016). The most popular examples of evolutionary radiations are probably adaptive radiations, in which the acquisition of a ‘key innovation’ spurs the rapid diversification of a clade accompanied by eco-morphological divergence (Gavrilets & Losos 2009; Glor 2010; Simpson 1953). This concept applies to a wide range of groups across the tree of Life. One may for instance refer to the angiosperms, of which the flower (along with other traits related to physiology, like the pace of living) is thought to have promoted their Late Cretaceous rapid diversification, Darwin’s ‘abominable mystery’ (Benton *et al.* 2022; Sauquet *et al.* 2017). Bats, the second most species-rich mammalian order, also stand out as an interesting example of adaptive radiation, their diversity being tightly related to the emergence of echolocation (Jones & Teeling 2006). Last, the radiation of hummingbirds (Aves: Trochilidae), accounting for nearly 400 species at present, can also be viewed as an example of adaptive radiation, enhanced by the acquisition of hovering flight and nectarivory (Mayr 2004). However, radiations are not always adaptive, and several works of the last few decades have called for terminological clarifications in that respect (Glor 2010; León-Tapia *et al.* 2026; Simões *et al.* 2016; Tu *et al.* 2025). The extant tree of life indeed offers striking examples of huge and rapid lineage accumulation that has not been accompanied by ecological divergence. This is for instance the case of Sigmodontinae (Rodentia: Cricetidae), the second most diversified rodent clade, and the most diversified mammalian group in the American tropics (León-Tapia *et al.* 2026; Maestri *et al.* 2017).

Furthermore, periods of diversity loss have long been recognised in the fossil record. These are well exemplified by the abrupt disappearance of trilobites or non-avian dinosaurs, respectively at the Permo–Triassic and the Cretaceous–Palaeogene boundaries. However, the biological relevance of these apparent extinction episodes has been called into question by the arising field of statistical palaeontology in the early 1970s (Sepkoski 2012). Two levels of concerns were expressed: (1) could these apparent extinction episodes in fact only reflect the incompleteness of the fossil record (Section 1.1.4) and (2) could these fluctuations differ from the predictions of a model in which speciation and extinction are purely random processes (*e.g.*, Gould *et al.* 1977; Raup *et al.* 1973; Raup 1978)? After a decade of methodological improvements, it has been established that ‘natural clades’ had features that could not be reproduced with simple non-mechanistic stochastic models (Sepkoski *et al.* 1981; Stanley *et al.* 1981). Coming along with external sources of validation (*e.g.*, the characterisation of an asteroid impact using Iridium anomalies at the precise time when dinosaurs disappeared from the fossil record (Alvarez *et al.* 1980)), five time frames with significantly elevated extinction levels over the Phanerozoic were thus characterised, formalised as the ‘big five mass extinctions’ (Raup & Sepkoski 1982) (depicted on Figure 1.4). Further research, first facilitated by the release of the high-resolution genus-level compendium of marine animals made by Sepkoski (2002), and then, by collective fossil data compilation efforts (*e.g.*, The Paleobiology Database) in fact transposed the definition of mass extinctions to other intervals of the Phanerozoic, stressing that these periods could show variable degrees of extinction magnitude (Bambach 2006; Hoyal Cuthill *et al.* 2020). This thesis will add some elements of discussion regarding the mass extinction status of one of these later-characterised mass extinction intervals, the Eocene–Oligocene Transition (see Chapter 1). That being said, mass extinctions arguably account for a minority of the extinctions that have occurred throughout the history of life (Quental & Marshall 2013). As stressed by Raup (1986), biodiversity is also affected by a “spectrum of smaller events” occurring at the scale of individual clades, or even lineages. Extinction

is thus not always abrupt: it can also operate gradually and, noteworthy, diversity losses occurring through the history of life may also be driven by decreases in speciation/origination rates (thus, not only by elevated extinction rates). This often overlooked perspective was termed the ‘Entwives effect’ by Quental & Marshall (2013), a metaphor from John R. R. Tolkien’s *Lord of the rings*, in which the Ents, a race of anthropomorphic trees, lost their wives, and thus had no means to have a descent (or to originate, in macroevolutionary terms).

Consequently, there is compelling evidence supporting that diversification rates varied across the history of life, and that these variations may be ruled by different evolutionary regimes (*e.g.*, gradual vs. mass extinction). It is now appropriate to move on to the factor(s) underlying these rate variations.

### Mode of Macroevolution: what drives rate variations?

The search for the mechanisms causing temporal changes in diversity – and thus diversification rates – has naturally come along with the first macroevolutionary reconstructions (Section 1.1.2). Noticing similar trends in the diversity curves of independent marine invertebrate clades, Newell (1952) immediately postulated the existence of “common causes” to which these clades must have been subject. These causes would “undoubtedly [be] complex and include both biological and physical factors” (Newell 1952). Nowadays, things have not gone farther that much: biodiversity is still considered as being affected by a mixture of biotic and abiotic factors (Benton 2009). Below is an attempt to produce a brief outline of the cornerstone rules that have been formulated to relate the pace of clade diversification to external drivers.

In 1973, Leigh Van Valen published a groundbreaking study that intrigued the community, starting from himself (Van Valen 1973). He evidenced that the fossil-derived survivorship curves of a wide range of taxa were mostly log-linear. This ‘Law of Constant Extinction’ implies that, at any time point in the evolutionary history of a clade, all the members of this clade are equally-likely to go extinct, irrespective of their age (Figure 1.7 B). This goes against alternative hypotheses instead stating that extinction rates should increase with taxon age (namely, age-dependent extinction, see Figure 1.7 A and C), prevailing at that time. Van Valen interpreted this finding from a Darwinian perspective. If extinction is indeed an age-independent process, the demise of a taxon could therefore only result from an ‘environmental deterioration’, and a subsequent failure of the taxon to keep up within its own ‘adaptive zone’ (in ecological terms), thus subsequently losing the ‘*struggle for life*’ (Darwin 1859). The extinction rate from the survivorship curve (Figure 1.7) would therefore provide a measure of the rate of environmental deterioration. Inspired by Lewis Carroll’s *Through the Looking-Glass*, when the Queen tells Alice that “*it takes all the running [she] can do, to keep in the same place*”, Van Valen thus named his *New Evolutionary Law* the *Red Queen* hypothesis (Liow *et al.* 2011).

The implications of the Red Queen hypothesis in terms of macroevolutionary modes lie within the way this ‘deteriorating environment’ is defined. Van Valen proposed that this constant environmental deterioration would result from the fact that all species within a given assemblages are part of a ‘zero-sum game’ driven by biotic interactions, mainly competition. This means that each minor gain of fitness for a given species would negatively impact the fitness of the others, but as they evolve as well, the potential compensations of fitness loss that could have been subsequently acquired would maintain the overall fitness of the system at a constant value. Consequently, the modern appreciation of the Red Queen hypothesis is that changes in diversification rates are mainly driven by biotic factors (*e.g.*, Barnosky 2001; Benton 2009).

Several studies exemplified that biotic interactions could affect clade diversification, thus providing



support for the Red Queen as primarily conceived by Van Valen. For instance, competition most likely shaped the macroevolutionary fate of canids (Silvestro *et al.* 2015a), sharks (Condamine *et al.* 2019b), antilocaprids (Nascimento & Pires 2025a), Cretaceous non-avian dinosaurs (Condamine *et al.* 2021) or gymnosperms (competing against angiosperms) (Condamine *et al.* 2020). Also, predation most likely affected the diversification of several prey and predator clades worldwide (*e.g.*, Nascimento *et al.* 2024; Nascimento & Pires 2025b; Prevosti *et al.* 2024; Solórzano *et al.* 2024; Tarquini *et al.* 2022). Another elegant example of how predation shapes clade diversification is the ‘*Escape and Radiate*’ model, proposed by Ehrlich & Raven (1964) and subsequently named by Thompson (1989). This model posits that phytophagous insects acquiring capacity to break the chemical defenses of a given plant clade undergo a rapid subsequent diversification (somewhat related to the notion of adaptive radiation, see previous sub-section) (Jousselin & Elias 2019). In response, the plant may evolve new chemical toxins, thus maintaining the selective pressure towards toxin-resistance among the insect clade (Jousselin & Elias 2019). Last, diversity-dependent diversification, namely, the set of hypotheses according to which diversification of a clade could be bounded by its own diversity (Rillo & Etienne 2022) can also provide insightful examples on how the Red Queen may shape the fate of the clades (*e.g.*, Condamine 2018; Condamine *et al.* 2019a; Etienne *et al.* 2012; Foote 2023).

Besides, although it clearly makes no doubt that the Red Queen’s predictions appropriately apply to a wide range of groups (as we intended to outline in the previous paragraph), it should be mentioned that Van Valen’s finding on age-independent extinction (Figure 1.7 B) triggered much debate in the palaeobiological community (Doran *et al.* 2006; Liow *et al.* 2011). Subsequent analyses of the fossil record of varied groups in fact often evidenced that groups could experience significant age-dependent extinction (ADE) patterns throughout their evolutionary history, preferentially towards younger species having greater extinction probabilities (*i.e.*, Figure 1.7 C) (*e.g.*, Condamine *et al.* 2019b; Finnegan *et al.* 2008; Kocáková *et al.* 2025; Saulsbury *et al.* 2024; Weppe *et al.* 2023), though noticeable studies also provided support for the opposite patterns (Figure 1.7 A) (*e.g.*, Guinot & Condamine 2023). The ADE must therefore be an additional example of the complexity of the biological reality and may selectively apply to some clades and not others. Also, as stressed by Hagen *et al.* (2018), one of the keys behind these apparent contradictory findings must lay within the taxonomic scale at which the study is conducted, most large-scale analyses of the fossil record histotically considering ranks above the species level, with ADE most likely not acting similarly at the level of species and higher taxa.

Another widely considered set of macroevolutionary hypotheses postulate that clade diversification may be impacted by phenotypical traits (Simpson 1953). The idea that phenotypical traits can influence diversification has indeed been around since the formulation of the concept of adaptive radiation, in which the brutal expansion of a clade intimately results from the emergence of a phenotypical feature (‘key innovation’, see previous sub-section) (Simões *et al.* 2016). A wealth of hypotheses have been formulated to relate clade diversification to phenotypical traits. For instance, larger body sizes have long been assumed to experience elevated extinction rates (Raup 1986). Also, pioneer work on insects showed that herbivorous lineages were more diversified than carnivorous lineages, suggesting diet-dependent diversification rates (Mitter *et al.* 1988). Subsequent methodological improvements led to the expansion of a class of diversification models known as State-dependent Speciation and Extinction (often abbreviated SSE), specifically investigating the existence of differential diversification regimes along trait categories (*reviewed in* Morlon *et al.* 2024), providing unprecedented insights into the mode of diversification for a wide range of groups. For instance, in the plant family Solanaceae, self-incompatible lineages were shown to diversify faster than self-compatible lineages (Goldberg *et al.* 2010). In Hymenoptera (wasps, ants, sawflies and bees), secondary phytophagy seems to be associated to elevated diversification rates (Blaimer *et al.* 2023). Also noteworthy, the methodological progress in

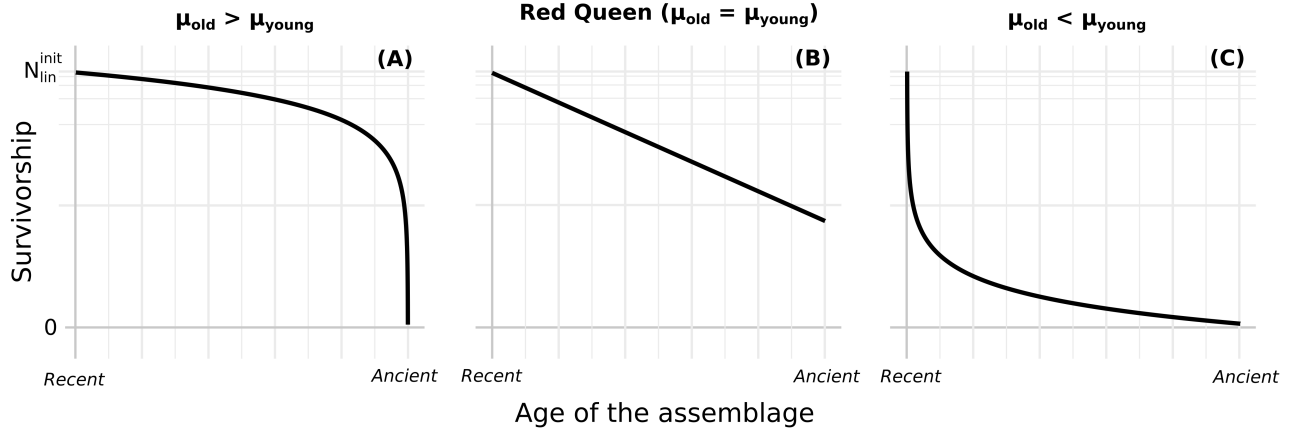


Figure 1.7: Expectations of survivorship as a function of time under three different scenarios. In all three representations, the slope of the curve represents the lineage loss per unit of time, and can thus be considered as a proxy of the extinction rate, noted  $\mu$ .  $N_{lin}^{init}$  depicts the initial number of lineages. In (A), extinction rates are higher in ancient times than in recent times, suggesting that older lineages have a higher extinction rate than younger lineages (hence,  $\mu_{old} > \mu_{young}$ ). The exact symmetrical case is shown in (C), where  $\mu_{young} > \mu_{old}$ . In (B), loss of lineages (hence extinction rates) does not vary across time. Extinction is thus age-independent, which is the primary evidence used by Van Valen to build the Red Queen hypothesis (Van Valen 1973). Note that the y axis has been log-scaled.

character evolution modelling (Beaulieu & O’Meara 2016) have allowed to invalidate several prevailing hypotheses regarding the effect of certain traits on diversification. Evidence from sharks (Marion *et al.* 2024) and amphibians (Liedtke *et al.* 2022) have indeed suggested that diversification was not related to reproduction mode in these clades, contrary to what was evidenced using earlier methods (e.g., Mull *et al.* 2024).

Alternatively, evidence primarily from the fossil record challenged the conspicuousness of the Red Queen as the main force driving diversification dynamics. Pioneering studies indeed emphasised alternative models in which Van Valen’s Law of Constant Extinction could in fact also arise from changes in species’ physical environment (e.g., Stenseth & Smith 1984; Vrba 1993). Thus, several works formulated hypotheses instead relating diversification dynamics to abiotic changes, stressing the occasional character of these physical disruptions (e.g., climatic hinge, bolide impact, volcanism). As an echo to Van Valen’s metaphor, these hypotheses were summarised under the *Court Jester* model by Barnosky (2001). The Court Jester hypotheses thus “share the common tenet that changes in the physical environment, rather than biotic interactions, are the initiators of major changes in organisms and ecosystems” (Barnosky 2001). Several empirical studies provided substantial support for such a model. Diversification dynamics of a wide range of clades were indeed found to be significantly impacted by temperature changes, for instance in butterflies (e.g., Condamine *et al.* 2018; Toussaint *et al.* 2025), ferns (e.g., Lehtonen *et al.* 2017), marine invertebrates (e.g., Guo *et al.* 2023; Malanoski *et al.* 2024), sharks (e.g., Condamine *et al.* 2019b) or mammals (e.g., Bekeraïté *et al.* 2025; Weppe *et al.* 2021). Mountain orogeny was also frequently illustrated among the main drivers behind clade diversification. The Himalayas indeed left significant imprints in the diversification of Apollo butterflies (Condamine *et al.* 2018) and plants (e.g., Pandit *et al.* 2025). Similarly, the build-up of the Andes affected the macroevolutionary fate of mammals (e.g., Pino *et al.* 2022; Solórzano *et al.* 2024;

Vallejos-Garrido *et al.* 2023), frogs and lizards (Boschman & Condamine 2022) or plants (Luebert & Weigend 2014). Additionally, major factors usually considered to impact clade diversification include palaeogeographical dynamics, with many studies supporting that view (*e.g.*, Malanoski *et al.* 2026).

As a result, it stands out that the modes of clade diversification are in fact most likely affected by a myriad of different biotic and abiotic factors (Benton 2009). The key would reside in a matter of scale (Gould 1985). Meanwhile the Red Queen is preferentially expected to occur at smaller spatio-temporal scales, the Court Jester would instead occur on longer timescales. Whether these scale differences affect the relative importance of one or the other in shaping the dynamics of clade diversification remains an open question.

### 1.2.2 Macro-eco-evolutionary understanding of the spatial organisation of life: insights from the Latitudinal Diversity Gradient

The uneven species richness across the globe that we outlined in Section 1.1.1, and notably, the striking pattern of increasing species richness with decreasing latitude famously known as the ‘Latitudinal Diversity Gradient’ (LDG), can be understood under a macroevolutionary perspective involving ecological processes. Several hypotheses (*reviewed in* Pontarp *et al.* 2019) have been proposed to explain such a pattern, some of which imply the interplay between diversification rates and tropical environmental conditions (Tietje *et al.* 2022).

In the footsteps of Stebbins (1974), the ‘cradle’ hypothesis posits that tropical lineages would have larger speciation rates than extra-tropical lineages. Higher speciation rates for tropical lineages may be predicted from tropic’s greater environmental heterogeneity (Stein *et al.* 2014), increasing available niche space. Conversely, one can expect speciation rates to be lower outside the tropics, where climate seasonality is more pronounced. In addition, the ‘Metabolic Theory of Ecology’ (Brown *et al.* 2004) can also predict that tropical lineages would have larger speciation rates. According to the latter, the higher energy availability for the tropics (higher temperature) would locally increase tropical species’ metabolic rate (Allen *et al.* 2002), yielding larger mutation, and thus, speciation rates (Allen *et al.* 2006; Lanfear *et al.* 2010). Also, as summarised by Mittelbach *et al.* (2007), tropical regions’ greater species richness has been suggested to enhance the importance of biotic interactions (Schemske *et al.* 2009). Consequently, since the ‘Red Queen’ hypothesis (Section 1.2.1) postulates that biotic interactions would enhance speciation, this would establish a positive feedback loop within the tropics, where increasing diversity would yield increasing speciation rates. This ‘cradle’ hypothesis is often contrasted with the ‘museum’ hypothesis, instead placing the emphasis on the fact that tropical lineages would have lower extinction rates. The ‘museum’ hypothesis may be a consequence of the ‘tropical stability’ hypothesis, assuming that the tropics have remained climatically stable across geological times, and that this stability would have made tropical lineages less prone to extinction (Wallace 1878). In addition, it might be expected that the aforementioned environmental heterogeneity offered by the tropics could buffer tropical lineages against eventual environmental changes.

A consensus emerged from late 2000s–2010s studies testing the extent to which diversification rates could explain the observed LDG, where the tropics may actually both act as a cradle and a museum of diversity – some works also emphasised the role of tropical-to-extratropical dispersal as a side cause behind the Great American Biotic Interchange, see Section 1.4.2. This view has been contrastingly supported in the recent literature. Nevertheless, joint analyses of fossil and phylogenetic data for several tetrapod clades, using cutting-edge methodological frameworks, provided support for elevated speciation rate outside the tropics as a likely mechanism for the LDG, thus instead favouring the ‘museum’ hypothesis (Meseguer & Condamine 2020; Quintero *et al.* 2023). Conversely, a recent



study focusing on a massive sample of the seed plant diversity emphasised that no causal relationship could be established between diversification rates and species richness (Tietje *et al.* 2022). Given these elements, it seems like the study model, the primary source of data (*e.g.*, fossils vs. dated phylogenies) and the methodology used to estimate diversification rates (see Section 1.3.1) affects upstream interpretations of the interplay between diversification rates and spatial patterns of species richness.

However, the notions of ‘cradle’ and ‘museum’ of biodiversity have recently been challenged by Vasconcelos *et al.* (2022), emphasising a number of limitations for the way they are often considered in the literature. Among them, as several factors driving speciation, such as climatic stability/heterogeneity, may also drive extinction, they argue that it would sometimes make more sense to consider turnover rates (see Glossary) rather than speciation and extinction rates individually. Also, they point out that extant centres of biodiversity with a great concentration of ‘basal’ lineages are not necessarily centres of origin, thus challenging both the way previous works actually interpreted the notion of museum and cradle of diversity. The latter nuance echoes previous works of the same authors emphasising the need to include hidden states in models of character evolution, marking a turning point in macroevolution (*e.g.*, Beaulieu & O’Meara 2016).

Alternative explanations of the LDG not involving diversification rates have also been proposed. These either refer to species’ ecological limits (*e.g.*, carrying capacity, size of the available area, palaeogeography) or time left for species to diversify (assuming tropical stability would have left more time for species to diversify within than outside the tropics, and attested by fossil and phylogenetic evidence of lineage age distribution in and outside the tropics) (Pontarp *et al.* 2019). As stressed by Mittelbach *et al.* (2007), these three classes of hypotheses (diversification-, ecological- or time-dependent) are not necessarily mutually exclusive. Conversely, a fourth set of hypotheses emphasised the likely implication of pure contingency in shaping the LDG, notably through the use of null models (Arita & Vázquez-Domínguez 2008). Recently, insights from eco-evolutionary simulations (see Section 1.3.2) evidenced that null models – in which the modelled patterns do not rely on particular eco-evolutionary mechanisms (Gotelli & McGill 2006) – could indeed match empirical LDG patterns, although generally worse at explaining empirical distributions data than model incorporating mechanistic constraints on species, in particular, climate and palaeogeography (Hagen *et al.* 2021a; Lorcery *et al.* 2025).

## 1.3 Analytical tools to study large-scale biodiversity patterns

This section is aimed to provide a broad outline of some methodological means to investigate the patterns and questions raised in the previous sections. Please note that each chapter of this thesis has its own *Methods* section.

### 1.3.1 Macroevoolutionary approaches to study how clades thrive and go extinct

#### The birth–death process

In macroevolution, the dynamics of clade diversification are usually modelled by a birth–death process (of course, this only holds for *model-based methods*, not for *count methods*, see next section) (see Tavaré 2018 for an exhaustive outlook). This stochastic process, borrowed from the field of population dynamics, is a classical example of continuous-time Markov process. Consequently, under this model, the number of species within a clade at any step  $i$  (here noted  $N_i$ ) only depends on the number of

species in the previous step ( $N_{i-1}$ ). In addition,  $N_i$  is ruled by two processes independently occurring at the scale of every individual lineages. Each lineage can indeed give birth to a new lineage (*speciation* event), or die (*extinction* event). In a birth–death process, speciation and extinction are modelled as Poisson point processes, and their respective intensities (also called rates) are usually noted  $\lambda$  and  $\mu$ , expressed in number of events per lineage per unit of time. Thus, the birth–death process makes the strong assumption that speciation and extinction are instantaneous events, which may not be appropriate, particularly in the case of speciation (see Morlon *et al.* 2024 for further discussion).

Kendall (1948) generalised the constant-rate birth–death process by allowing  $\lambda$  and  $\mu$  to be functions of time. Half a century later, this foundational mathematical development spurred significant theoretical and computational innovations in the field of macroevolution (again, see Morlon *et al.* 2024 of an extensive review). Notable improvements of the birth–death process for the inference of clade diversification have involved the incorporation of fossil and extant sampling parameters (e.g., Heath *et al.* 2014; Stadler 2010), a fossilisation parameter (e.g., Silvestro *et al.* 2014b; Stadler *et al.* 2018); have allowed for more precise estimations of the timing of diversification rate variations (e.g., Kopperud & Höhna 2025; Quintero *et al.* 2024; Rabosky 2014; Silvestro *et al.* 2019); state-dependent diversification rate variations (e.g., Beaulieu & O’Meara 2016; Maddison *et al.* 2007); rate variations between clades (Morlon *et al.* 2011) or even lineages (Maliot *et al.* 2019; Quintero *et al.* 2024); rate variations along external abiotic and/or biotic correlates (e.g., Etienne *et al.* 2012; Lehtonen *et al.* 2017; Licht *et al.* 2026; Silvestro *et al.* 2015a) or combinations of several of these features (e.g., Hauffe *et al.* 2024; Quintero *et al.* 2023).

## Quick overview of the philosophy behind the Bayesian approach to parameter inference

Today, some diversification models – and notably those applied over this thesis – are implemented in a Bayesian framework. It is thus appropriate to clarify some key notions of Bayesian inference before moving on to the models *per se*.

Let’s say one wants to explain some data  $\mathcal{D}$  using a statistical model  $\mathcal{M}$ , going along a **set parameters**  $\theta$ . The type of data is not important here, but to make it more anchored in the overall framing of this doctoral work, it can be considered that  $\mathcal{D}$  is a set of fossil occurrence data (defined by their age), and that  $\mathcal{M}$  is a diversification model, thus including two key parameters (origination and extinction rates, respectively  $\lambda$  and  $\mu$ ), so that  $\theta = \{\lambda, \mu\}$  (note that  $\theta$  may include any additional parameter **related any** other process incorporated within  $\mathcal{M}$ , such as sampling, see previous section).

The goal of any model-based statistical inference is to explain  $\mathcal{D}$  with  $\mathcal{M}$ . This aim may be achieved with two **kind** of approaches implying fundamentally distinct ways of reasoning. The first one consists of finding the set of parameter values  $\theta^*$  maximising the probability of the data given the model, also called *likelihood*, noted  $\mathcal{P}(\mathcal{D}|\theta)$ . This is typically done in *Maximum-Likelihood* inference, and results in a single estimate of parameter values ( $\theta^*$ ), also called point estimate. Conversely, the initial goal may also be fulfilled by focusing on the probability of the model given the data, called *posterior probability* and noted  $\mathcal{P}(\theta|\mathcal{D})$ , which is the core of *Bayesian* inference. The Bayes’ theorem links the likelihood and the posterior probability in the following way:  $\mathcal{P}(\theta|\mathcal{D}) = \frac{\mathcal{P}(\mathcal{D}|\theta)\mathcal{P}(\theta)}{\mathcal{P}(\mathcal{D})}$ .

$\mathcal{P}(\theta)$  is called *prior probability* (often also simply referred to as ‘prior’). It reflects potential *a priori* knowledge on model parameters, which constitutes one of the strengths of the Bayesian approach. In Bayesian inference, it is indeed possible for the user to inform the model, and thus, to sort of guide the parameter inference step. That being said, priors have also been the subject of many of the critics formulated against Bayesian inference, since they may also be viewed as a way to bias the inference

step. The case of informative *vs.* non-informative priors has also been much debated (Lemoine 2019), but will not be detailed here.

Also,  $\mathcal{P}(\mathcal{D})$  represents the *marginal probability* of the data. It could be obtained by integrating over all possible parameter combinations ( $\mathcal{P}(\mathcal{D}) = \int_{\theta} \mathcal{P}(\mathcal{D}|\theta)d\theta$ ), but this is often computationally hard to achieve in the real world. To circumvent this apparent hindrance to posterior probability calculation, Markov Chain Monte Carlo (MCMC) algorithms are used during Bayesian parameter inference, relying on computing posterior ratios between two sampling steps within the parameter space, thus avoiding to calculate marginal probabilities (see Barido-Sottani *et al.* 2024 for clear in-depth explanations).

Overall, one of the main forces of Bayesian inference is that it estimates parameter distributions, thus offering a direct quantification of the uncertainty around parameter estimation, strikingly contrasting with Maximum-Likelihood’s point estimates – requiring bootstrapping to get an estimate of uncertainty.

### Fossil occurrence-based methods to infer the dynamics of clade diversification

The first estimations of origination and extinction rates in macroevolution have relied on the only direct source of evidence of life in the past, that is, the fossil record (*e.g.*, Newell 1952). These may be grouped into two different classes.

*Count methods* have been the first historically applied. As their name indicates, these have originally consisted of approximating origination and extinction rates within a given time bin with raw counts of the number of taxa appearing or disappearing within such a bin (*e.g.*, Newell 1952; Webb 1976). However, the palaeobiological community’s growing concern about the caveats of the fossil record (Section 1.1.4) challenged the accuracy of such estimates. Notably, the characterisation of the ‘pull of the recent’ and the ‘Signor-Lipps’ effects (Section 1.1.4) implied to re-think the structure of these inherently biased sampling schemes, triggering substantial methodological improvements within that set of methods. Several of them have consisted of (1) relying on consecutive time windows sliding throughout the lifespan of a given taxon and (2) applying some criteria indicating whether or not such a taxon should be incorporated within a given bin with respect to the other windows. Bin-specific origination and extinction rates were thus computed based on equations developed out of these adapted counting schemes (reviewed in Alroy 2014). The most popular of these methods include the ‘Boundary Crosser’ (Foote 2000), the ‘two-timer’ and ‘three-timer’ (Alroy 2008) or the ‘Gap-Filler’ (Alroy 2014) equations.

On the other hand, over the past ten years, *model-based methods* relying on a birth–death process (see previous sub-section) have been an increasingly more frequent means to estimate diversification rates from fossil occurrences. To account for some of the key biases of the fossil record, this set of methods may explicitly incorporate those biases as probabilistic processes. For instance, PyRate (Silvestro *et al.* 2014a), probably the most popular of these frameworks (and intensively used over this thesis, see Chapter 1 and 2) **jointly infers diversity, origination and extinction rates along with a preservation process, explicitly modelling fossilisation bias** (Silvestro *et al.* 2014b). In addition, PyRate offers the possibility to account for (1) potential lineage-heterogeneous preservation intensities (Section 1.1.4) by means of a gamma model – just like for substitution rates in phylogenetic inference – and (2) the time-heterogeneity of the preservation process by allowing preservation rates to vary in a set of user-defined time bins (Section 1.1.4) (Silvestro *et al.* 2019). In addition, PyRate has been implemented in a Bayesian framework, thus offering a direct access to parameter uncertainty (see previous section).

PyRate has known substantial improvements throughout the past 10 years, that allowed to test for

increasingly complex macroevolutionary hypotheses. Notable progress have for instance involved the possibility to assess for the effect of environmental variables on rate variations (Lehtonen *et al.* 2017; Silvestro *et al.* 2015a) or to test for age-dependent extinction (see Section 1.2.1) (Hagen *et al.* 2018). More recently, using deep-learning, the Birth–Death Neural Network (BDNN) model has extended PyRate’s framework by allowing for lineage-time-specific diversification rates estimated as non-linear combinations of a user-defined set of environmental predictors, thus significantly expanding the set of testable macroevolutionary scenarios (Hauffe *et al.* 2024).

However, PyRate’s main shortfall is that it makes the assumption of complete lineage sampling and therefore does not incorporate sampling bias in its framework. This is problematic since empirical datasets, and particularly fossil occurrences, are subject to high sampling heterogeneities (*cf.* Section 1.1.4). Methodological efforts have been invested to account for this limitation. Some include applying spatial standardisation procedures to the input dataset and/or incorporate a Bernoulli process at the base of the diversity inference to account for unsampled lineages (Flannery-Sutherland *et al.* 2022). More recently, training a deep-learning algorithm with a large number of simulated fossil datasets has provided promising results for leveraging the impact of sampling bias on taxon richness across time (Cooper *et al.* 2024). Also, other occurrence-based diversification models exist, leveraging the aforementioned incomplete sampling shortfall, such as the treeless Fossilised Birth–Death range model (Stadler *et al.* 2018; Warnock *et al.* 2020), explicitly incorporating a sampling-through-time parameter (Stadler 2010).

### Phylogeny-based methods to infer the dynamics of clade diversification

*As this thesis will not apply such methods, only a brief outline will be given here.*

Phylogenies are commonly-used representations of the relationships between a given set of taxa by means of acyclic undirected graphs (‘phylogenetic trees’), in which terminal nodes (‘leaves’, at the end of the terminal edges, ‘branches’) represent the taxa, and the initial node (‘root’) represents the common ancestor to each of these taxa. Every internal node has the particularity of being dichotomous, and thus, is connected by exactly three branches. Dated phylogenies, in which branch lengths are proportional to divergence times between species, have long been recognised to contain precious information about the dynamics of clade diversification (Morlon 2014). With a simple lineage-based model in which, at every time step, lineages could speciate or go extinct with given probabilities, Raup *et al.* (1973) simulated phylogenies with impressively similar features to empirical trees. In fact, without explicitly naming it, they demonstrated that it was possible to model the process of diversification with a birth–death model (see previous sub-sections), which was further formalised two decades later (Nee *et al.* 1994). From these outstanding contributions, a wide array of increasingly-complex methods have been developed, reviewed in Morlon *et al.* (2024).

A subtle point of the approach lies in that, in macroevolution, the birth–death process is the preferred model to build trees (other tree-building methods exist, such as the coalescent, but their use is not common in macroevolution, see Cornuault *et al.* (2025)). On the other hand, birth–death models may also be fitted to already-reconstructed dated trees to make some inferences on the most likely diversification processes explaining the tree, for instance, using model selection (*e.g.*, Condamine *et al.* 2019a; Lewitus *et al.* 2018). Conceptual issues may arise from this subtlety, for instance, when fitting birth–death models with time-variable rates to trees reconstructed under a constant-rate birth–death process.

It should also be mentioned that estimating diversification rates from phylogenies has been inten-

sively debated in the community. Particularly, several studies raised the issue that the shape of extant phylogenies could be consistent with an infinity of speciation and extinction rate combinations, and thus, that these two parameters were not identifiable (the problem coming from extinction rate estimation using extant-only phylogenies, greatly subsampling the evolutionary history of the focal clade) (Louca & Pennell 2020; Rabosky 2010). One solution to circumvent such a critical bias lies in the combination of extant and fossil data, for instance, in an approach known as the Total-Evidence Dating (Gavryushkina *et al.* 2017; Pyron 2011; Warnock & Wright 2020). This powerful framework relying on a tree-building process known as the Fossilised Birth–Death (FBD) process (Mulvey *et al.* 2025) was indeed shown to leverage the aforementioned identifiability issue (Truman *et al.* 2024). However, such models are computationally intensive. In addition, the two sources of information used by the FBD applied to Total-Evidence Dating (molecules and morphology, but **not** that the FBD process may be used to carry out phylogenetic reconstructions from molecular-only datasets) are sometimes difficult to reconcile as not necessarily subject to the same evolutionary constraints. Also, these two sources of data are not always accessible, *e.g.*, in groups with no known fossil record. Consequently, although it stands out as an ideal avenue to infer diversification from phylogenies, such a framework may be hard to apply in practice.

### 1.3.2 Integrating ecological processes with evolutionary forces: mechanistic eco-evolutionary simulations

The traditional approach to eco-evolutionary modelling has consisted of fitting models to data (as seen for diversification models, see Section 1.3.1). Such an approach is of course inherently affected by the existing biases within the data, intensively discussed in the previous sections (Section 1.1.4), and has left little room for casual testing of the actual mechanisms underlying the observed patterns. The use of simulation models has opened worthy avenues to overcome these issues (Gotelli *et al.* 2009). Among them, spatially-explicit population-based mechanistic eco-evolutionary models (MEEMs) have proven highly relevant to tackle issues related to the emergence and the dynamics of biodiversity patterns across deep time (Hagen 2023).

MEEMs combine a gridded landscape made up with a set of user-selected variables (*e.g.*, palaeoclimate) with a set of eco-evolutionary rules to simulate virtual species. The use of simulated data allows MEEMs to leverage any caveat coming along with empirical data. Since the very first introduction of simulation-based models in ecology and evolution by Gotelli *et al.* (2009), several MEEMs have been released, offering unprecedented insights into the mechanisms at play to fashion the spatio-temporal dynamics of biodiversity (*e.g.*, Leprieur *et al.* 2016; Rangel *et al.* 2018; Saupe *et al.* 2020; Skeels *et al.* 2023b). One of the most popular is the general engine for eco-evolutionary simulations, also called *gen3sis*, that has been used over this thesis (Chapter 3)(Hagen *et al.* 2021a).

In *gen3sis*, virtual species are modelled with individual populations (one population per grid) evolving independently in the pre-defined landscape, following a set of four rules. (1. *Dispersal*) Populations may disperse from one cell to another according to a user-defined dispersal kernel (often a Weibull distribution). (2. *Speciation*) If two populations happen to be disconnected, as long as they remain so, they will accumulate divergence at a user-defined rate across simulations steps. If inter-population divergence falls above a user-defined speciation threshold, these two populations will convert into two distinct species. Consequently, the only speciation mode that *gen3sis* is so far able to simulate is allopatry (see Glossary). (3. *Environmental filtering*) Species may be given environmental tolerances (or niches) along the variables included in the landscape, determining their level of fit to local environment at the cell scale. (4. *Trait Evolution*) Species-specific niche are defined following key traits



(*e.g.*, temperature optimum, breadth, competitive ability) that may themselves evolve across simulation steps, or be fixed. Trait evolution may be driven by environmental changes, thus offering the possibility to test for different ecological scenarios (null or neutral vs. adaptive).

Consequently, by directly specifying the eco-evolutionary processes mediating species' ability to maintain within a landscape, gen3sis (as well as any other MEEM) stands out as a powerful and flexible tool to mechanistically assess a wide range of hypotheses on how biodiversity patterns emerge and are maintained over time.

## 1.4 A strange biodiversity on a strange land : the case of mammals in South America

*"[The Neotropical] region includes South America, the Antilles and tropical North America; and it possesses more peculiar families of vertebrates and genera of birds and mammalia than any other region"* (Wallace 1876)

### 1.4.1 South America across the Cenozoic: elements of palaeogeographical and palaeoenvironmental context

Due to its unique physical geography, South America offers an immense diversity of habitats, from tropical alluvial plains, dry highlands, grasslands to the top of the second highest mountains on Earth (Figure 1.8). Understanding the build-up of South American landscapes is vital to disentangle the build-up of its astonishing biodiversity.

South America had a very singular palaeogeographic history. Akin to Australia in the present-day, the continent spent most of the Cenozoic completely isolated from other landmasses (Scotese & Wright 2018). In terms of faunal evolution, such an isolation had strong implications, that will be further detailed in Section 1.4.2.

Probably the main feature that stands out as one stares at the present-day map of South America is its marked topographical heterogeneity. The eastern side of the continent indeed appears topographically homogeneous, despite the existence of a few rolling areas like the Brazilian Highlands or the Guianian Shield (both composed of billion-years-old cratons, Figure 1.8). On the contrary, with the Andes stretching North-to-South over 8000 km, the western margin of South America is considerably more topographically heterogeneous. The Andes, including more than 70 peaks above 5000 m and two out of the three highest plateaus on Earth, are the result of the subduction of the Nazca plate beneath the South American plate (Ramos 2009). Although the Andean subduction system started way earlier, the Andean orogenetic (*i.e.*, mountain-building) cycle began in the late Early – early Late Cretaceous (Horton 2018; Ramos 2009). This uplift intensified from the Late Miocene onwards, showing a pronounced northwards-propagating diachrony (Boschman 2021; Gianni *et al.* 2020) (Figure 1.10). The eastern part of the Andes (matching the Andean foreland basins, Figure 1.8), offering a wide altitudinal range and a tropical climate, is arguably the most species-rich region in the world (Figure 1.2), and the origin of its huge local diversity has puzzled biogeographers since the times of Humboldt (Antonelli *et al.* 2018; Rahbek *et al.* 2019a, b). In addition, it should be mentioned that a significant amount of low-latitude fossil evidence unearthed in the recent years came from this foreland basin (Antoine *et al.* 2017) (Figure 1.6, 1.8 and 1.9).

The palaeogeographical, palaeoclimatic and palaeoenvironmental history of South America during the Cenozoic was tightly related to the pace of the Andean uplift and its spatial dynamics (Hoorn *et al.* 2010). A wealth of evidence indeed attests that the elevation of this mountain chain profoundly re-organised continental climate, notably since it represents a topographic barrier to atmospheric circulation, and thus, aridified some places and intensified precipitations in others (Blisniuk *et al.* 2005; Hartley 2003). These effects, of increasing intensity as the Andes got higher (notably since the Middle Miocene, Figure 1.10), combined with global-scale temperature dynamics greatly shaped the landscapes at the continent scale, and notably the rise of open landscapes at the expense of closed forests (*e.g.*, Palazzesi & Barreda 2012).

The Andean uplift also exerted a major control on continent-scale hydrographic patterns, in turn strongly impacting the structure and composition of faunal and floral assemblages locally. From the Early to the late Middle Miocene (*ca.* 20–10 Ma), a vast wetland known as the ‘Pebas MegaWetland System’ (PMWS, Figure 1.10) covered Western Amazonia, occupying significant portions of what is now Colombia, Ecuador, Peru, Venezuela and Brazil (Hoorn *et al.* 2022a). There is clear evidence supporting that, together with eustatism, the Andean uplift was the primary control of the emergence and sustaining of this system, mostly by causing subsidence and locally increasing precipitation rates (Hoorn *et al.* 2022a and literature therein). The PMWS was characterised by a complex mixture of interconnected aquatic environments, such as lakes or swamps (somewhat analogous to what may be found in the Pantanal region in Brazil, see Marivaux *et al.* (2020a) and Hoorn *et al.* (2022b)), and was home to a unique biodiversity, very distinct from what is recovered from extant assemblages in the region (*e.g.*, Antoine *et al.* 2017; Salas-Gismondi *et al.* 2015; Tejada-Lara *et al.* 2015). The demise of the PMWS was most likely also a by-product of the Andean uplift. Similarly, there is strong evidence supporting that the Late Miocene marine introgression that flooded parts of what is now Northern Argentina, Brazil and Southern Bolivia, known as the ‘Paranense sea’ (named after the region of Paraná, see Figure 1.8 and 1.10) was primarily controlled by the Andean uplift (Hernández *et al.* 2005). The ecosystems generated by the Late Miocene demise of the Paranense sea are suspected to have been favoured lineage diversification, as shown for caviomorph rodents (*e.g.*, Candela *et al.* 2012).

Last, South American ecosystems were also immensely impacted by the closure of the isthmus of Panama, regarded as an incontestable facilitator for the inter-American faunal admixture known as the ‘Great American Biotic Interchange’ (GABI, see Figure 1.10 and Section 1.4.2) (Jaramillo 2018). The timing of the completion of an overland connection between the Americas, having profound implications for the temporal calibration of the GABI, has been widely debated. The age of 3.5 Ma has long been used as a benchmark (Bacon *et al.* 2015) but geological (Coates & Stallard 2013; Montes *et al.* 2015) and biological (Bacon *et al.* 2015) evidence also suggested an earlier age, raising some criticism (Lessios 2015; Marko *et al.* 2015; O’Dea *et al.* 2016). Nowadays, integrating data from several areas of Earth and Life sciences resulted in a consensus in which deep-water flow in the isthmian region ceased with the closure of the Central American Seaway by the late Middle Miocene (*ca.* 10 Ma). After this date, it is generally accepted that the Pacific and the Caribbean still exchanged shallow water until the Late Pliocene (*ca.* 3.5–2.8 Ma), marking the emergence of a land bridge – the so-called isthmus of Panama – that most likely persisted until nowadays (Jaramillo 2018; O’Dea *et al.* 2016).



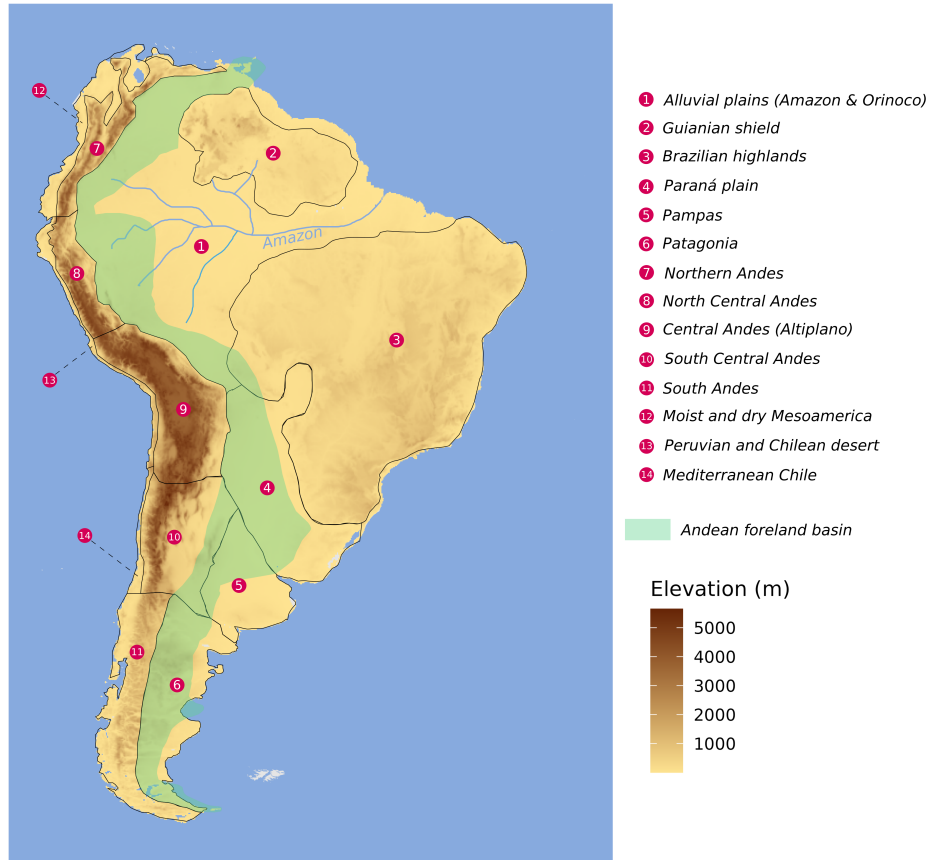


Figure 1.8: Extant physiography of the South American continent. 14 among the main (eco-) physical regions of the continents were represented on top of the present-day topography of the continent, from Boschman (2021). The approximate contour of the Andean foreland basin has been highlighted in light green.

#### 1.4.2 'Splendid isolation': the captivating history of South American mammals across the Cenozoic

##### History of mammal palaeontology in South America: a brief overview

*Some of the historical elements developed here may be found in further detail in [Manias \(2023\)](#).*

South American mammal palaeontology sadly started as a colonisation by-product, as European explorators plundered Patagonian soils, intrigued by skeletal remains trapped in the rocks. In the late 18<sup>th</sup> century, some of the fossils brought back to Europe provoked curiosity among the leading figures of the emerging field of comparative anatomy. As a result, in 1796, Georges Cuvier described the famous *Megatherium americanum*, and proposed this six-meter-beast to be a sloth. A few decades later, in 1836, Charles Darwin, back from his five-years journey on board the HMS Beagle, brought back to England a wealth of fossil material (in particular from the region of Punta Alta, Patagonia – Figure 1.9). These fossil remains were given to the renowned palaeontologist Richard Owen, who described, in 1837, many emblematic fossil mammals from South America, including the car-sized

armadillo *Glyptodon*, the giant sloth *Mylodon*, the giant camel-like native ungulate *Macrauchenia* or the rhino-sized native ungulate *Toxodon*.

The second half of the 19<sup>th</sup> century marked the beginning of South American mammal palaeontology made by scientists from South America, or more precisely, from the Republic of Argentina. At that time, natural sciences, as well as any other field contributing to sharpening knowledge on the country's own territory, were envisaged as nation-building, and the Southern part of the country (Pampas and Patagonia provinces) had long been recognised to overflow with fossil remains of extinct animals. Consequently, important efforts were devoted to collect and shed light on the country's archaeological and palaeontological patrimony, centralised within national museums, such as the “Museo Nacional de La Plata”, headed by the prominent explorer and natural scientist Francisco Moreno in the 1870s.

South American mammal palaeontology owes a lot to the labour of the **patriotic** Argentinian palaeontologist Florentino Ameghino. Relying on fossil material mostly collected during field expeditions led by his brother Carlos, Florentino Ameghino produced a remarkable quantity of systematic palaeontological work, and his written productions are regarded as essential building-blocks for South American palaeomammalogy. These are probably best exemplified by his foundational monographies, such as “*Contribución al conocimiento de los mamíferos fósiles de la República Argentina*” (Ameghino 1889), in which several hundreds of taxa from the Argentinian fossil record are described. Argentina's fossil richness, together with the aforementioned efforts to enlighten its natural patrimony, made that South American palaeomammalogy long remained essentially constrained to the country, hence the striking spatial clustering of the fossil occurrences in the Southern cone depicted on Figure 1.9. As recalled by Simpson (1940), in the first half of the 20<sup>th</sup> century, knowledge of Cenozoic mammal remains outside Argentina was very scarce. Many of the most significant discoveries of non-Argentinian mammal-bearing fossil localities (as those highlighted on Figure 1.9) have indeed occurred throughout the last 80 years, with notable exceptions of long-recognised rich fossil-bearing localities, such as Tarija, in Southern Bolivia (Boule 1920).

In the 1930s, the young U.S. vertebrate palaeontologist George G. Simpson, later to become the renowned architect of the Modern Synthesis of Evolution (Futuyma 2015), carried out his first expeditions in South America, intrigued by the continent's peculiar fossils. Simpson further developed a strong interest towards understanding the evolution of South American mammals, that animated him all over his career (e.g., Simpson 1940, 1950, 1967; 1970). In 1980, he produced an outstanding synthesis of South American mammalian evolution across the Cenozoic, that is still used as a reference today (Simpson 1980). In his book, he proposed that the tempo of South American mammalian evolution could be summarised by three temporally-distinct ‘faunal strata’, that will be briefly outlined in the subsequent sections (Figure 1.10)👉

### First stratum: endemic radiations (Late Cretaceous–middle Eocene)

Simpson's first stratum characterises the early phase of South American mammal evolution in the aftermath of the continent's isolation (Figure 1.10). It is thought to have lasted from the Late Cretaceous/early Palaeocene to the late middle Eocene. Mammal at that time are often referred to as the continent's ‘original inhabitants’. These would include endemic metatherians (Figure 1.11 A), native ungulates (Figure 1.11 C), xenarthrans (Figure 1.11 B), as well as a few remnant non-therian lineages (e.g., Gondwanatheria).

The oldest Cenozoic South American land mammalian assemblages date back to the early Palaeocene. These have been recovered from the **Punta Peligro (Argentina)** and **Tiupampa (Bolivia)**

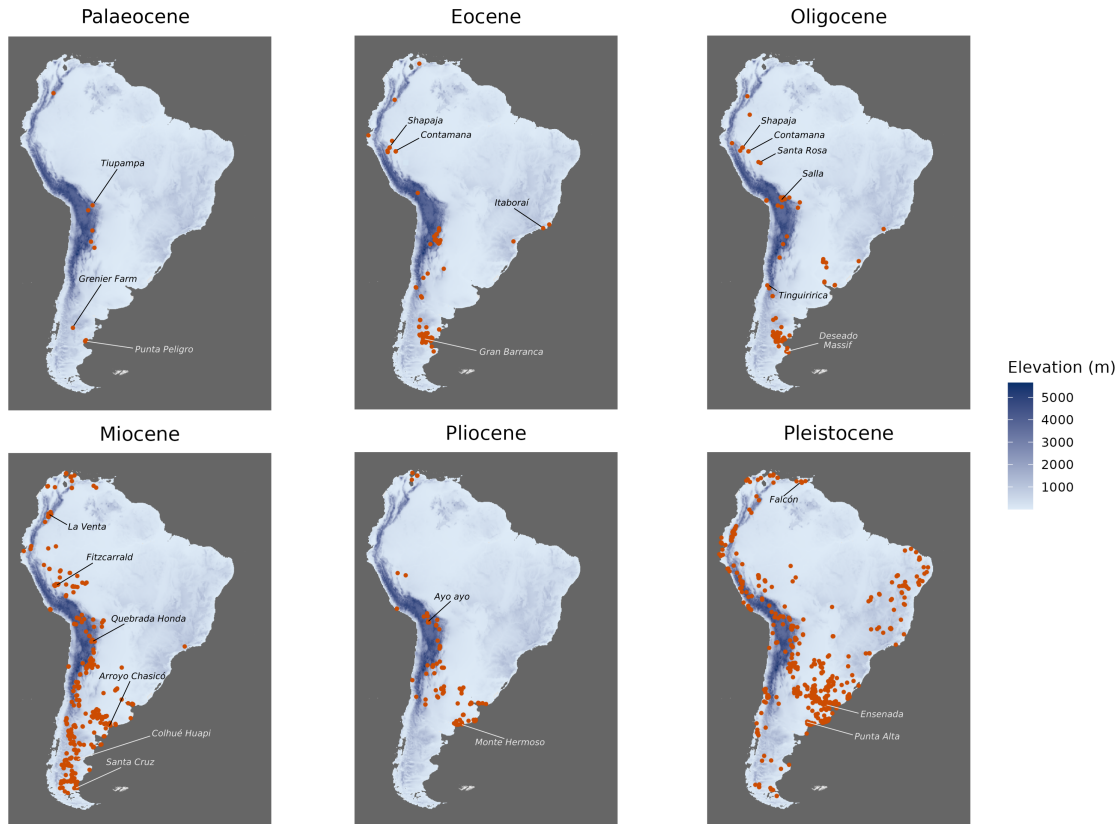


Figure 1.9: Distribution of the South American terrestrial mammal fossil record across geological epochs (Holocene excluded), on top of the continent's present-day topography (Boschman 2021). Several keystone mammal-bearing localities have been highlighted.

localities (Figure 1.9), each leading to the recognition of the oldest intervals of the South American Land Mammal Age (SALMA) scale, namely, the Peligran and the Tiupampan (Figure 1.10). Biochronological evidence suggests the Tiupampan SALMA to be older than the Peligran SALMA, and corroborate an age interval of 64.5–64 Ma for the Tiupampan fauna (Gelfo *et al.* 2009). An additional occurrence of pre-Tiupampan therian mammal was recovered from the Grenier Farm locality (Figure 1.9), and was assigned to the extinct metatherian order Polydolopimorphia, Figure 1.11.

The faunal composition of the Tiupampan and Peligran assemblages delivered precious insights into the early Cenozoic history of South American mammals. The great metatherian diversity (Figure 1.11 A), together with the presence of representatives from several ungulate-related lineages typically found in the Northern hemisphere (formerly grouped into the ‘condylarths’), suggested that South American mammalian assemblages received significant faunal supply from North America between the late Cretaceous and the early Palaeocene (Pascual 2006). Geological evidence corroborated the possibility for intermediate episodic connections between the two continents in that interval (Pascual 2006), possibly also involving the Caribbean islands (Viñola-López *et al.* 2022). As an echo to the name ‘Great American Biotic Interchange’ (see Section 1.4.2 and Chapter 3), this keystone biotic exchange for the architecture of Simpson’s first stratum was thus termed ‘First American Biotic Interchange’ (FABI) (Goin *et al.* 2012a).

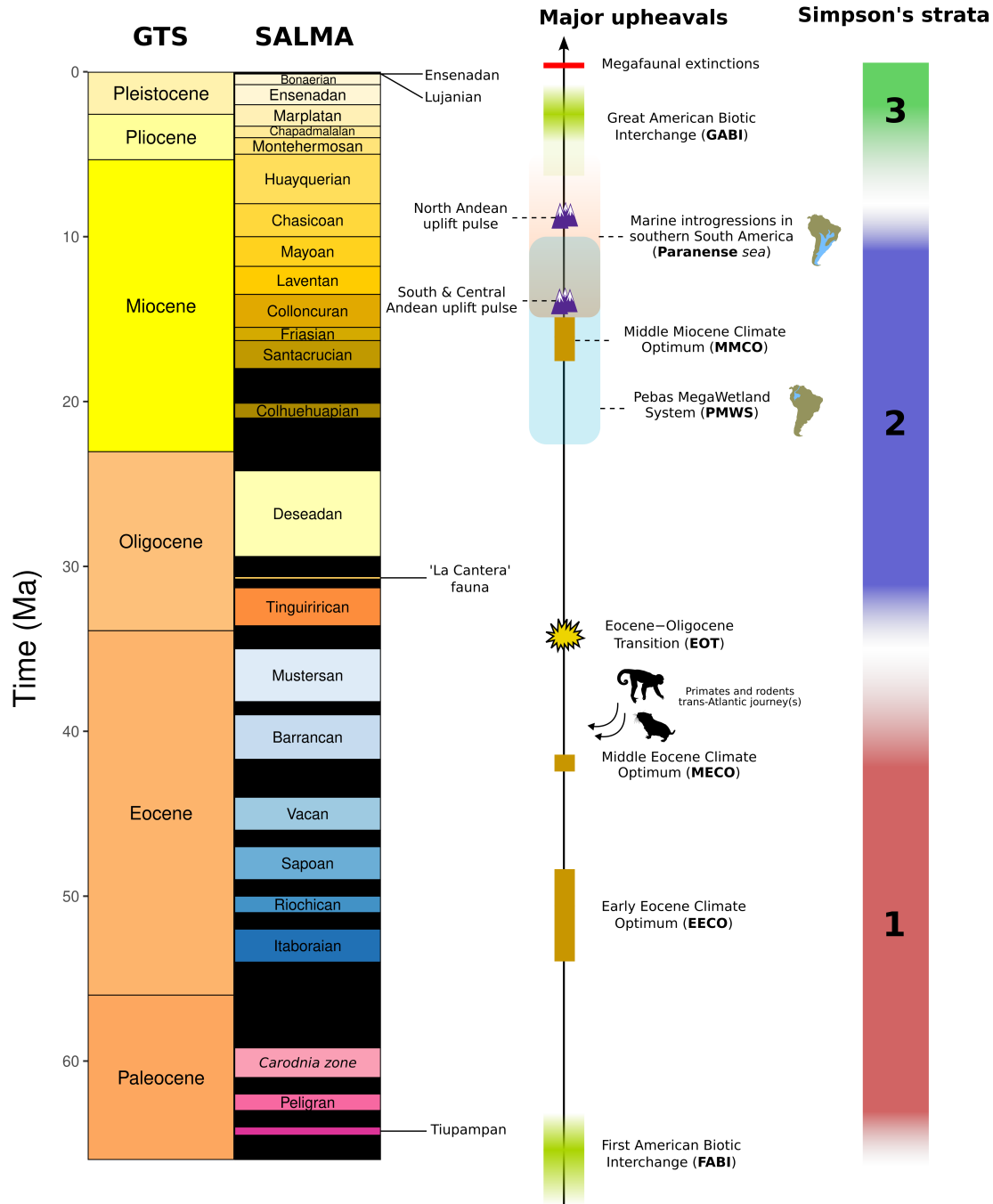
Along with these therian lineages of North American affinity, non-therian lineages persisted in South America at least until the mid-Palaeogene, thus well after the Cretaceous–Palaeogene boundary (Goin *et al.* 2012b). The most fascinating example of non-therian lineage persistence known to date is probably the enigmatic *Necrolestes*, from the Santa Cruz Formation (early Miocene, Santacrucian SALMA, see Figure 1.10), identified as a remnant of the clade Meridiolestida, a mammal group that thrived from the Late Jurassic up until the Late Cretaceous (Chimento *et al.* 2012; Rougier *et al.* 2012).

Younger assemblages characteristic of Simpson’s first stratum notably include the São José de Itaboraí (often abbreviated Itaboraí) fauna (Figure 1.9), one of the most diverse in the Palaeogene of South America. It led to the Itaboraian SALMA, approximately dating back to 53 Ma (Figure 1.10). The Itaboraian fauna is still dominated by metatherians (Goin *et al.* 2016). It also revealed that all the higher South American Native Ungulate (SANU) lineages were clearly differentiated at that time (Figure 1.11 C). More importantly, the Itaboraian fauna yielded the oldest known occurrence of armadillo (Xenarthra: Cingulata) (Bergqvist *et al.* 2004), thus documenting the appearance of Xenarthra in the fossil record (Figure 1.11 B). The late appearance of xenarthrans in the fossil record remains an enigma for South American mammal palaeontology, and has been referred to as the ‘xenarthran mystery’ (Goin *et al.* 2016). Indeed, as suggested by their name meaning “weird articulations”, xenarthrans, today represented by sloths, anteaters and armadillos, share very derived morphological characters that are not found in any other known eutherian mammal lineage. Molecular evidence suggests that they form a clade with the African group Afrotheria, called ‘Atlantogenata’, literally meaning “generated by the Atlantic ocean”, and thus, would have diverged from their sister-group following the break-up of Gondwana, some 120 Ma (Waddell *et al.* 1999). Why armadillos only appear in the fossil record *ca.* 53 Ma, and sloths in the Late Oligocene (see next section), whereas their divergence with other South American eutherians largely predates the earliest known mammal assemblages found on the continent thus remains an open question.

Younger assemblages of the first stratum progressively record a turnover in the relative proportion of metatherian vs. eutherian mammals in the communities, with an increasing dominance of eutherians, that several authors related to the gradual Eocene climate cooling from *ca.* 50 Ma onwards (Croft 2016; Goin *et al.* 2016; Westerhold *et al.* 2020). Later in the Eocene, land mammal assemblages will know go through a series of gradual changes that will progressively mark the transition with the subsequent faunal stratum, detailed thereafter.

## Second stratum: Trans-atlantic migrations and faunal ‘modernisation’ (late Eocene–Late Miocene)

Most specialists agree that major re-organisations occurred within South American mammal assemblages near the Eocene–Oligocene boundary (that will be more thoroughly covered in Chapter 1), marking the onset of Simpson’s second faunal stratum (Figure 1.10). The distinction between the second and the first strata lies in a three-part faunal ‘modernisation’ (Flynn & Wyss 1998): (1) emergence of morphological innovations among “original inhabitants”, (2) extinction of basal, ‘archaic’ lineages and (3) higher-level taxonomic origination within “original inhabitants”, alongside the arrival and radiation of hystricognathous rodents (Rodentia: Hystricognathi: Caviomorpha) and anthropoid primates (Primates: Anthropoidea) in South America (Figure 1.10).



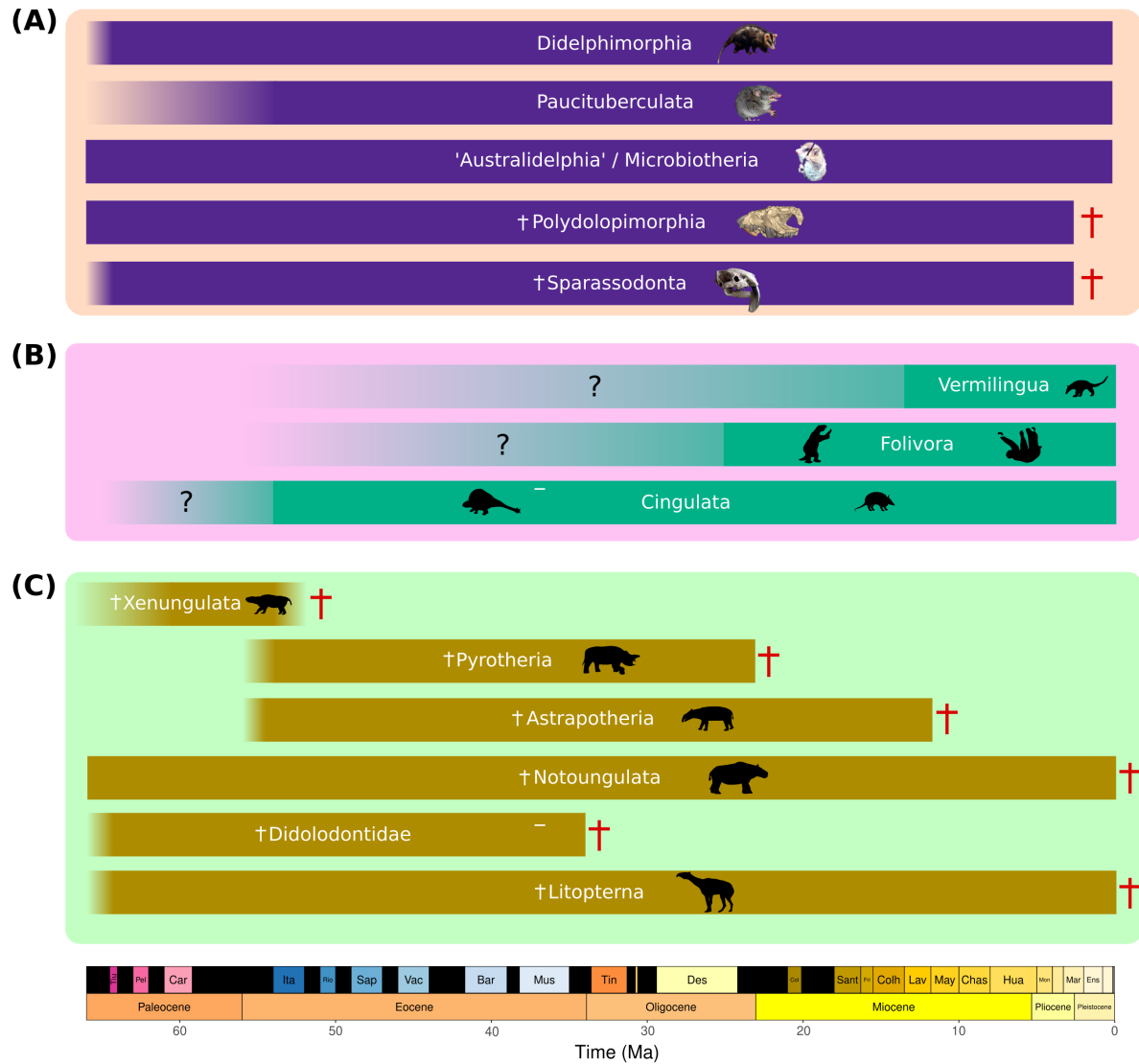
**Figure 1.10:** Integrative view of the Cenozoic of South America from the land mammal perspective. The South American Land Mammal Age (SALMA) scale used all over this thesis is shown, along with the corresponding Geological Time Scale (GTS) epochs, from the [International Chronostratigraphic Chart](#) (v2024/12) (Holocene not depicted). Boundaries between SALMAs were defined on the basis of the most up-to-date literature (*e.g.*, [Bellinzoni et al. 2025](#); [Dunn et al. 2013](#); [Gelfo et al. 2009](#); [Woodburne et al. 2014](#)), but note that many intervals lack absolute temporal calibration. Black spaces between SALMAs represent temporal gaps. Alongside are depicted several key biotic or abiotic upheaval in the Cenozoic history of South American mammals, as well as Simpson's three faunal strata ([Simpson 1980](#)). Silhouettes are from Phylopic (credit: Sarah Werning and Kai R. Caspar).

The most notable morphological innovation that could be mentioned is probably the increase in cheek teeth crown height (*i.e.*, hypsodonty), occurring independently among several (at least six) groups of native ungulates (particularly notoungulates) from the middle Eocene onwards (Croft *et al.* 2020). The evolution towards hypsodonty in South America is particularly interesting as hypsodont cheek teeth started to be recorded 20 million years before the first reports of hypsodonty outside South America (García-López *et al.* 2019). As they diversified, freshly-arrived caviomorph rodents (see later in this section) also, for some of them, rapidly acquired hypsodont cheek teeth, again, way before any other place in the world (late Oligocene–Early Miocene), as exemplified by the superfamily Chinchilloidea – see Chapter 2. The Eocene emergence of hypsodont cheek teeth and their Oligocene spread across lineages has long puzzled South American mammal palaeontologists (*e.g.*, Billet *et al.* 2009; Croft *et al.* 2003; Reguero *et al.* 2008, 2010). Hypsodont cheek teeth are indeed often considered as an indicator of an abrasive diet, typically, incorporating grasses, and thus, have often been related to landscape opening (Stebbins 1981). However, palaeoenvironmental evidence failed to support any concomitant grassland spreading in the regions where the first hypsodont lineages were recorded (*e.g.*, Kohn *et al.* 2004, 2015; Strömberg *et al.* 2013). Now, a consensus emerged in the community, as several lines of evidence instead suggested that the **increase in cheek teeth crown height may have instead resulted from grit or volcanic dust consumption** (Armella & Croft 2026; Damuth & Janis 2011; Jardine *et al.* 2012; Selkin *et al.* 2015).

Besides, by analysing two early Oligocene metatherian communities from Southern South America (one of Tinguirican age, the other being the La Cantera fauna, see Figure 1.10), Goin (2010) reported an important taxonomic turnover occurring among metatherians in the few million years that followed the Eocene–Oligocene transition (EOT). This geological boundary, that will be further explored in Chapter 1, has been identified as the most intense Cenozoic climatic disruption, as global temperature cooled down by several degrees in a few hundred thousands of years (Coxall & Pearson 2007; Liu *et al.* 2009; Westerhold *et al.* 2020). **The EOT was related to a major faunal turnover among mammalian communities from the Northern hemisphere** (Meng & McKenna 1998; Stehlin 1909; Stucky 2014). As an echo to the french expression ‘Grande Coupure’ (Stehlin 1909) used to characterise the mammalian turnover that occurred among European mammal assemblages, Goin (2010) thus derived the expression ‘Bisagra Patagónica’ (*Patagonian hinge*) to qualify the faunal turnover that they characterised.

Next, as we briefly evoked, Simpson’s second stratum is also characterised by the abrupt apparition of hystricognathous rodents and anthropoid primates in the South American fossil record (Antoine *et al.* 2012; Bond *et al.* 2015; Marivaux *et al.* 2023; Seiffert *et al.* 2020). Explaining such an emergence has been one of the most important challenges for South American mammal palaeontology, given that, contrary to xenarthrans (see section of stratum 1), it made no doubt that these groups were the result of a much more recent faunal supply (Simpson 1950). After several decades of intense debate, combined morphological (Hoffstetter 1972) and molecular (*e.g.*, Poux *et al.* 2006) evidence provided strong support for the “rafting hypothesis”, in which both groups would **descend** from Afro-Asian ancestors that would have crossed the Atlantic in the Palaeogene (*e.g.*, Antoine *et al.* 2012; Marivaux *et al.* 2023) (Figure 1.10).





**Figure 1.11:** Higher-level persistence of South American mammals from Simpson's first stratum in the fossil record. **(A)** Metatherians, at the order level. Fading colours mean that the group has no known fossil record for the period, but should already exist. Photos were taken from iNaturalist (credit: Jorge Brito, Eduardo Silva Rodriguez, Thiago Chagas). *Wamradolops telloi* cranium comes from Stutz *et al.* (2026), reproduced with permission. The picture of the *Thylacosmilus atrox* cranium was taken by Claire Houck. **(B)** Xenarthra. Fading colours with question marks depict the 'xenarthran mystery', *i.e.*, that the group in question has no known fossil record for the period, but should already exist for the highlighted interval. Divergence ages between the main lineages were taken from Tejada *et al.* (2024). Vermilingua and Folivora altogether form the order Pilosa. Silhouettes are from Phylopic (credit: Julián Bayona and Sarah Werning). **(C)** South American Native Ungulates (SANUs), adapted from Croft *et al.* (2020). Fading colours depict uncertainty in the age of earliest fossil. Silhouettes are from Phylopic (credit: Julián Bayona).

Despite its wide acceptance, the timing of this fascinating trans-Atlantic journey remains under debate. Seiffert *et al.* (2020) proposed that the global sea-water drop that followed the peri-EOT climate cooling, by reducing the distance between Africa and South America, could have increased the likelihood for **a trans-oceanic dispersal** (assuming the latter to be inversely proportional to the distance between the source and the receiving areas). Such a hypothesis finds support in that, according to a study conducted by the same authors, the oldest fossil evidence for rodents and primates in South America cannot be older than the early Oligocene (Campbell *et al.* 2021). Conversely, Marivaux *et al.* (2023) proposed that this trans-Atlantic dispersal may have instead occurred 5 million years earlier, near the Middle Eocene Climate Optimum (MECO) (Bohaty *et al.* 2009). This interval may have been accompanied by intense meteorological episodes in the tropics (Mulch *et al.* 2015), increasing the likelihood of riverbank break-ups, and thus, the formation of floating islands serving as a means for the large-scale dispersal of small-sized mammals, such as rodents or primates (Houle 1998). This hypothesis more adequately explains the high taxonomic diversity displayed by several known early Oligocene rodent communities, accounting for taxa clearly identified as crown caviomorphs, with recognised members of Chinchilloidea (*e.g.*, Antoine *et al.* 2021; Arnal *et al.* 2022 *though some assemblages listed here may be of latest Eocene age*; Bertrand *et al.* 2012; Boivin *et al.* 2018; 2019). Such a diversity would be hard to achieve in the few hundred thousand years separating these assemblages from the EOT. Note that, given the paucity of the primate fossil record in the Palaeogene of South America – only four specimens assigned to three distantly-related lineages that notably, and contrary to rodents, are not recognised as stem representatives of the extant South American primate radiation, namely, Platyrrhini (Marivaux *et al.* 2023) – it is currently impossible to apply such a limitation to the primate scenario proposed by Seiffert *et al.* (2020).

In addition, the early Oligocene interval has been recognised as the most likely time frame for the colonisation of the West Indies by several taxa, including crown caviomorph rodents (Marivaux *et al.* 2020b), sloths (Tejada *et al.* 2024) as well as other terrestrial taxa (Blackburn *et al.* 2020; Chamberland *et al.* 2018). These colonisations would have been favoured by the emergence of an ephemeral land bridge between the Greater Antilles and the South American landmass – the Aves Ridge –, following the aforementioned sea-level drop after the EOT, a hypothesis called GAARlandia (Cornée *et al.* 2021; Iturralde-Vinent & MacPhee 1999; Philippon *et al.* 2020). However, it should be mentioned that such a hypothesis has raised criticism (*e.g.*, Ali 2012; Ali & Vences 2019).

Simpson's second stratum persisted through the Oligocene and the Miocene. Keystone late Oligocene localities, such as Salla (Bolivia) or the Deseado massif (Argentina) (Figure 1.9) provided unprecedented insight into South American mammal diversity at that time. In particular, they helped identify the first records of sloths (Pujos & De Iuliis 2007) as well as to better constraint the diversification of Notoungulata, Litopterna, Astrapotheria and the extinction of Pyrotheria near the Oligocene–Miocene (Croft *et al.* 2020) (Figure 1.11 C). Then, from the early Miocene, sampling gets increasingly less temporally discontinuous (Figure 1.10), thus better constraining our knowledge of the evolution of mammals on the continent. In addition, molecular and fossil data agree that most of the crown extant South American mammal families that survived until the present diverged in the Miocene (Álvarez-Carretero *et al.* 2022). Thus, mammal diversity from the Late Oligocene to the Late Miocene was flourishing, and many invaluable sites or geological formations (*e.g.*, Santa Cruz Formation, La Venta, Quebrada Honda,... see Figure 1.9 as well as Croft (2016)) yielded precious discoveries attesting that statement.

### Third stratum: Inter-american exchange, massive faunal disruptions, and establishment of the extant mammal biodiversity (latest Miocene–Present)

Please note that, all-over this manuscript, “North America” will actually refer to North and Central America, roughly, everything above the Isthmus of Panama.

The flourishing diversity that characterised Simpson’s second stratum was massively disrupted in the late Neogene. As seen in Section 1.4.1, ca. 3 Ma, a sustainable landbridge connecting the Americas emerged, known as the Isthmus of Panama. Though noticeable examples of earlier colonisations have been documented in the fossil record (Woodburne 2010), it is consensual to consider that the closure of the isthmus of Panama catalysed the admixture between North and South America’s endemic mammalian faunas, an event famously known as the ‘Great American Biotic Interchange’ (GABI) (Webb 1985) (Figure 1.10).

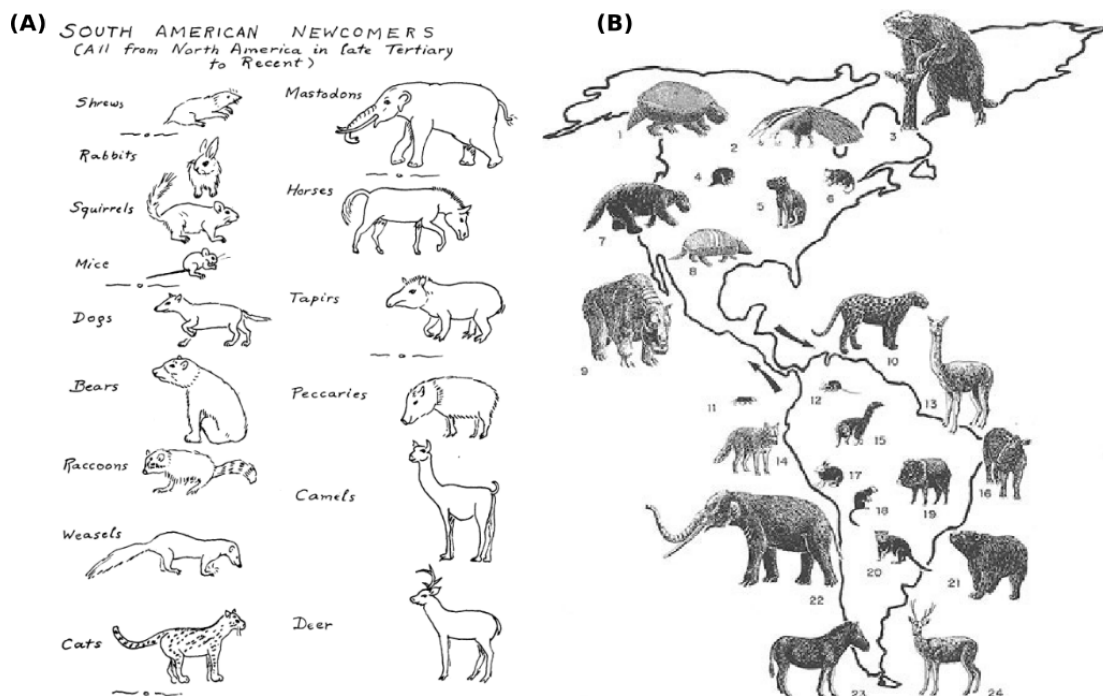


Figure 1.12: (A) Hand-drawing of the North American taxa recorded in the fossil record of South America, hence considered ‘newcomers’, made by Simpson (1950). (B) Schematic interpretation of land mammalian exchange across the Americas following the GABI, from (Vrba 1992). All the taxa shown in North America are of South American origin, and vice-versa. *In North America*: 1. glyptodon, 2. anteater, 3. and 7. ground sloths, 4. New-world porcupine, 5. capybara, 6. opossum, 8. armadillo, 9. toxodontid notoungulate. *In South America*: 11. shrew, 12. sigmodontine cricetid, 13. camelid, 14. canid, 15. mustelid, 16. tapir, 17. rabbit, 18. squirrel, 19. peccary, 20. racoon, 21. bear, 22. gomphotheriid, 23. horse, 24. cervid.

As described by Woodburne (2010), although the early phases of the GABI were characterised by roughly symmetrical exchanges (as illustrated by the various groups on Figure 1.12 B), during the Pleistocene, the fossil record suggests that an asymmetry of exchange gradually established, with North American taxa colonising South America with a higher success than the other way around.

At present, it is estimated that around 50% of South American mammal's generic diversity descend from a North American lineage, whereas this proportion falls down to some 10% when considering genera with a South American ascent living in North America (Webb 1991). The latter proportion is considerably reduced to a few single species when excluding tropical North America, a region that even Wallace recognised as “the portion of the North American continent which belongs zoologically to South America” (Wallace 1876). Consequently, it stands out that the GABI has been highly asymmetric.

What shaped the striking asymmetry of this lineage exchange has been one of the most puzzling issues in South American mammal palaeontology. Several hypotheses have indeed been proposed to explain such a exchange imbalance, that will be intensively reviewed and discussed in Chapter 3. The first hypotheses formulated involved some kind of competitive superiority of North American taxa out of South Americans (*e.g.*, Simpson 1950). Later hypotheses instead placed the emphasis on the role of palaeoclimate change (*e.g.*, Vrba 1992; Webb 1991). Simpson (1980) even termed such an exchange a ‘fortuitous experiment’, arguing that if things were to be repeated, the resulting pattern would thus not necessarily be equivalent. On top of this, confounding effects may arise from megafaunal extinctions (Hadler *et al.* 2026; Svenning *et al.* 2024), several studies arguing that hunting may also have caused the demise of endemic (large-sized) south American clades in South America (Boscaini *et al.* 2025; Prates & Perez 2021). Consequently, despite being the subject of a wide range of studies, the drivers behind the asymmetry of the GABI remain an open question, and one of the core aims of the present doctoral work is to contribute to a better understanding of this fascinating enigma.

### 1.4.3 Challenges for South American mammal palaeontology

Despite being unique and invaluable, several challenges arise as we intend to work with the South American mammal fossil record, in addition to those previously mentioned arising while using fossils (Section 1.1.4).

As it shows on the maps displayed in Figure 1.9, the fossil record of South American mammals is strikingly uneven across space, with the persistence of a huge gap merely overlapping with the plain of the Amazon river, the Brazilian Highlands and the Guianian Shield (Figure 1.8). Almost 30 years ago, Flynn & Wyss (1998) already pointed the need to fill the pronounced gap in the tropical fossil record of South American mammals, being highly sparse, with the notable exception of the exceptional Miocene fauna from the La Venta locality, Colombia (Carrillo *et al.* 2023) (Figure 1.9). This is primarily the result from three factors. First, except along the Andes where the subduction has actively promoted sediment accumulation in the Andean foreland basin (Figure 1.8), tropical South American soils experience little subsidence, and consequently, being less dense than continental crust, hardly can sediments get buried in the Amazonian plain (and thus offering the possibility for fossils to form). Second, tropical soils are markedly more active in terms of microfauna. Decomposition there is thus more efficient than outside the tropics, reducing the probability for a dead body to fossilise. Last, obvious accessibility issues hamper fossil-hunting in the tropical rainforest. Without invoking geopolitical factors, low-latitude fossil-bearing localities from Western Amazonia are indeed almost all located nearby brand-new roads across the forest or along rivers, posing additional logistic issues (Figure 1.13).

Substantial improvements should however be stressed, notably in the Peruvian Amazonia, where exceptional fossil deposits have been found (*reviewed in* Antoine *et al.* 2017). These are greatly exemplified by the regions of Contamana and Shapaja, where nearly 100 different vertebrate-bearing localities spanning distinct intervals of the Cenozoic have been characterised (Antoine *et al.* 2016;

Antoine *et al.* 2021), including CTA-27, where the oldest caviomorph rodent community of the continent was unearthed (Antoine *et al.* 2012). Significant contributions should also be mentioned from the early Oligocene locality of Santa Rosa (Campbell *et al.* 2004), which yielded two out of the three **basalmost** primates found in South America to date (Bond *et al.* 2015; Seiffert *et al.* 2020). Despite all these costly efforts, critical time intervals (such as the middle-late Oligocene or the early Miocene) remain unsampled under tropical latitudes, superimposing on the existing SALMA scale’s gaps (depicted in black on Figure 1.10). Consequently, every palaeontologist who ever worked in tropical South America knows this ‘South American mammal fossil record paradox’, in which the region encompassing the greatest diversity at the scale of the continent (and even the world, see Figure 1.2) is also, by far, the most fossil-poor.

Such an incompleteness was shown to significantly affect upstream macroevolutionary analyses relying on the South American mammal fossil record (Ugarte *et al.* 2025), and should thus be kept in mind **throughout the rest of this dissertation**.



Figure 1.13: Example of a fossil **locality along the Tambo river**, Peruvian Amazonia (photo by the author).

## 1.5 **Objectives** of this doctoral work

This thesis aims to improve the understanding of key **moments** in South American mammal evolutionary history through an interdisciplinary approach of macroevolution, macroecology and palaeobiology.

First, emphasis will be placed on the **Eocene–Oligocene interval** (*ca.* 56–23 Ma), genuinely straddling Simpson’s first and second strata, and encompassing the most dramatic Cenozoic climate disruption (the Eocene–Oligocene Transition, EOT, *ca.* 34 Ma). Though the EOT has been associated to a marked mammal turnover (sometimes even considered as a mass extinction) in the Northern Hemisphere, continent-scale evidence in South America is lacking. The first chapter of this thesis (**Chapter 1**) will contribute to the discussion on the status of the EOT as a mass extinction by **reconstructing** the Eocene–Oligocene diversification dynamics of South American mammals, relying on an unprecedented dataset and cutting-edge preservation-aware methods.



Throughout the course of the first chapter, long-standing issues about diversification regimes under the tropics raised since the times of Wallace (Section 1.2.2) will be approached. Specifically, whether tropical and extratropical clades exhibit similar or distinct macroevolutionary trajectories will be explored, along with the latitudinal (un)conspicuousness of the environmental forces underlying the reconstructed diversification trajectories.

Next, this thesis will also contribute to a better understanding of the emergence of singular phenotypic features among South American land mammals. By reconstructing the trait- and environment-dependent diversification history of Chinchilloidea (Rodentia: Caviomorpha), a rodent clade poorly diverse at the present with a rich fossil record encompassing the largest rodents to ever fool the ground of the Earth, foundational hypotheses on trait-dependent diversification (Section 1.2.1) will be explored. Ultimately some light will be shed on the existence of differential macroevolutionary tempo between chinchilloid body size and hypsodonty categories, meanwhile pinpointing the likely environmental drivers underlying the diversification dynamics of this emblematic clade. Also, the first chapter of this thesis will provide quantitative insights into how grassland spreading shaped South American mammal diversification dynamics in the Palaeogene, which has historically been invoked as the driver behind the strikingly earlier emergence of hypsodonty in mammals from South America compared to the rest of the world, but further largely criticised.

Also, in its last chapter, this thesis will provide valuable mechanistic insights into what shaped a textbook example of asymmetric lineage exchange following a geodispersal event, namely, the Great American Biotic Interchange (GABI) (Section 1.4.2). The imbalanced success of North American mammal lineages compared to South Americans in the aftermath of the closure of the isthmus of Panama has indeed yielded to a wide range of hypotheses. By means of computer experiments, this thesis will specifically assess whether such an asymmetry can be linked to palaeoclimate dynamics, a long-considered paradigm.

The findings produced over this thesis will ultimately be replaced in a broader framework, discussing the likely triggers of macroevolutionary change among South American mammals, and several perspectives will be provided to enhance further macro-eco-evolutionary reflexion on the fate of this fascinating study model.